

Elektrodynamik Formelsamling

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1 Math

1.1 Coordinates

Cartesian Coordinates

$$d\mathbf{l} = dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}}, \quad d\tau = dx dy dz$$

Gradient:

$$\nabla t = \frac{\partial t}{\partial x} \hat{\mathbf{x}} + \frac{\partial t}{\partial y} \hat{\mathbf{y}} + \frac{\partial t}{\partial z} \hat{\mathbf{z}}$$

Divergence:

$$\nabla \cdot \mathbf{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$$

Curl:

$$\nabla \times \mathbf{v} = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \hat{\mathbf{z}}$$

Laplacian:

$$\nabla^2 t = \frac{\partial^2 t}{\partial x^2} + \frac{\partial^2 t}{\partial y^2} + \frac{\partial^2 t}{\partial z^2}$$

Spherical Coordinates

$$d\mathbf{l} = dr \hat{\mathbf{r}} + r d\theta \hat{\boldsymbol{\theta}} + r \sin \theta d\phi \hat{\boldsymbol{\phi}}, \quad d\tau = r^2 \sin \theta dr d\theta d\phi$$

Gradient:

$$\nabla t = \frac{\partial t}{\partial r} \hat{\mathbf{r}} + \frac{1}{r} \frac{\partial t}{\partial \theta} \hat{\boldsymbol{\theta}} + \frac{1}{r \sin \theta} \frac{\partial t}{\partial \phi} \hat{\boldsymbol{\phi}}$$

Divergence:

$$\nabla \cdot \mathbf{v} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 v_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta v_\theta) + \frac{1}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi}$$

Curl:

$$\nabla \times \mathbf{v} = \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (v_\phi \sin \theta) - \frac{\partial v_\theta}{\partial \phi} \right] \hat{\mathbf{r}} + \left[\frac{1}{r \sin \theta} \frac{\partial v_r}{\partial \phi} - \frac{1}{r} \frac{\partial}{\partial r} (r v_\phi) \right] \hat{\boldsymbol{\theta}} + \left[\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) - \frac{1}{r} \frac{\partial v_r}{\partial \theta} \right] \hat{\boldsymbol{\phi}}$$

Laplacian:

$$\nabla^2 t = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial t}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial t}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 t}{\partial \phi^2}$$

Cylindrical Coordinates

$$d\mathbf{l} = ds \hat{\mathbf{s}} + s d\phi \hat{\boldsymbol{\phi}} + dz \hat{\mathbf{z}}, \quad d\tau = s ds d\phi dz$$

Gradient:

$$\nabla t = \frac{\partial t}{\partial s} \hat{\mathbf{s}} + \frac{1}{s} \frac{\partial t}{\partial \phi} \hat{\boldsymbol{\phi}} + \frac{\partial t}{\partial z} \hat{\mathbf{z}}$$

Divergence:

$$\nabla \cdot \mathbf{v} = \frac{1}{s} \frac{\partial}{\partial s} (s v_s) + \frac{1}{s} \frac{\partial v_\phi}{\partial \phi} + \frac{\partial v_z}{\partial z}$$

Curl:

$$\nabla \times \mathbf{v} = \left[\frac{1}{s} \frac{\partial v_z}{\partial \phi} - \frac{\partial v_\phi}{\partial z} \right] \hat{\mathbf{s}} + \left[\frac{\partial v_s}{\partial z} - \frac{\partial v_z}{\partial s} \right] \hat{\boldsymbol{\phi}} + \frac{1}{s} \left[\frac{\partial}{\partial s}(s v_\phi) - \frac{\partial v_s}{\partial \phi} \right] \hat{\mathbf{z}}$$

Laplacian:

$$\nabla^2 t = \frac{1}{s} \frac{\partial}{\partial s} \left(s \frac{\partial t}{\partial s} \right) + \frac{1}{s^2} \frac{\partial^2 t}{\partial \phi^2} + \frac{\partial^2 t}{\partial z^2}$$

1.2 Vectors

$$a(\mathbf{A} + \mathbf{B}) = a\mathbf{A} + a\mathbf{B}$$

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}| \cos(\theta) \quad (1.1)$$

$$\mathbf{A} \times \mathbf{B} = |\mathbf{A}||\mathbf{B}| \sin(\theta) \hat{\mathbf{n}} \quad (1.4)$$

$$(\mathbf{A} \times \mathbf{B}) = -(\mathbf{B} \times \mathbf{A}) \quad (1.6)$$

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}) \quad (1.15)$$

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) \quad (1.17)$$

$$(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{C} \times \mathbf{D}) = (\mathbf{A} \cdot \mathbf{C})(\mathbf{B} \cdot \mathbf{D}) - (\mathbf{A} \cdot \mathbf{D})(\mathbf{B} \cdot \mathbf{C})$$

$$\mathbf{A} \times [\mathbf{B} \times (\mathbf{C} \times \mathbf{D})] = \mathbf{B}[\mathbf{A} \cdot (\mathbf{C} \times \mathbf{D})] - (\mathbf{A} \cdot \mathbf{B})(\mathbf{C} \cdot \mathbf{D}) \quad (1.18)$$

1.3 Calculus

$$dT = \left(\frac{\partial T}{\partial x} \hat{\mathbf{x}} + \frac{\partial T}{\partial y} \hat{\mathbf{y}} + \frac{\partial T}{\partial z} \hat{\mathbf{z}} \right) \cdot (dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}})$$

$$= (\nabla T) \cdot d\mathbf{l}, \quad (1.35)$$

where the gradient (∇) is an operator:

$$\nabla = \hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z}. \quad (1.39)$$

The gradient is used to calculate the divergence of a vector with:

$$\nabla \cdot \mathbf{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}, \quad (1.40)$$

and the curl with:

$$\nabla \times \mathbf{v} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix}$$

$$= \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \hat{\mathbf{z}}. \quad (1.41)$$

The product rules for the gradient are:

$$\begin{aligned}
\nabla(fg) &= f\nabla g + g\nabla f \\
\nabla(\mathbf{A} \cdot \mathbf{B}) &= \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A} \\
\nabla \cdot (f\mathbf{A}) &= f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f) \\
\nabla \cdot (\mathbf{A} \times \mathbf{B}) &= \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}) \\
\nabla \times (f\mathbf{A}) &= f(\nabla \times \mathbf{A}) + \nabla f \times \mathbf{A} \\
\nabla \times (\mathbf{A} \times \mathbf{B}) &= \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}) + (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B}
\end{aligned}$$

When we use the gradient to make a second derivative we get the Laplacian:

$$\nabla(\nabla T) = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}, \quad (1.42)$$

and the laplacian of a vector:

$$\nabla^2 \mathbf{v} = (\nabla^2 v_x)\hat{\mathbf{x}} + (\nabla^2 v_y)\hat{\mathbf{y}} + (\nabla^2 v_z)\hat{\mathbf{z}}. \quad (1.43)$$

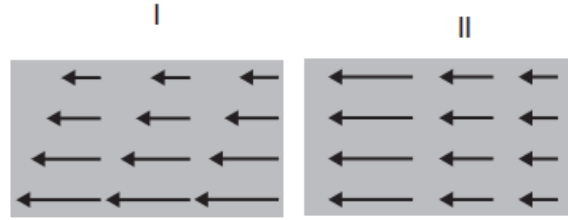
We also have the following identities:

$$\nabla \times (\nabla T) = 0 \quad (1.45)$$

$$\nabla \cdot (\nabla \times \mathbf{v}) = 0 \quad (1.46)$$

$$\nabla \times (\nabla \times \mathbf{v}) = \nabla(\nabla \cdot \mathbf{v}) - \nabla^2 \mathbf{v}. \quad (1.47)$$

In terms of vector fields, the field is divergent if the vectors change "speed" coming in and out of frame. The curl means the arrows rotate, or that the "speed" is different between the top and bottom of the frame. In the figure below I has curl and no divergence, and II has divergence but no curl.



Dirac Delta Function

$$\begin{aligned}\delta(x-a) &= \begin{cases} 0, & \text{if } x \neq a \\ \infty, & \text{if } x = a \end{cases} \\ \int_{-\infty}^{\infty} \delta(x-a) dx &= 1 \\ \int_{-\infty}^{\infty} f(x) \delta(x-a) dx &= f(a) \\ \int_{-\infty}^{\infty} f(x) \delta(g(x)) dx &= \sum_{x_i} \frac{f(x_i)}{|g'(x_i)|} \\ \delta(x^2 - a^2) &= \frac{\delta(x-a) + \delta(x+a)}{2|a|}\end{aligned}$$

$$\nabla \cdot \left(\frac{\hat{\mathbf{z}}}{z^2} \right) = 4\pi\delta^3(\mathbf{z}), \quad (1.100)$$

$$\nabla \left(\frac{1}{z} \right) = -\frac{\hat{\mathbf{z}}}{z^2}, \quad (1.101)$$

$$\nabla^2 \frac{1}{z} = -4\pi\delta^3(\mathbf{z}). \quad (1.102)$$

1.4 Fundamental theorem of calculus

Gradient:

$$\int_a^b (\nabla T) \cdot d\mathbf{l} = T(\mathbf{b}) - T(\mathbf{a}) \quad (1.55)$$

Divergence:

$$\int_V (\nabla \cdot \mathbf{v}) dV = \oint_S \mathbf{v} \cdot d\mathbf{a} \quad (1.56)$$

Curl:

$$\int_S (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \oint_{\mathcal{P}} \mathbf{v} \cdot d\mathbf{l} \quad (1.57)$$

2 Electrostatics

2.1 Electric field

The force on a test charge (Q) from a single point charge (q), a distance $\mathbf{z}$ away, is calculated by Coulomb's law:

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{z^2} \hat{\mathbf{z}}, \quad (2.1)$$

where $\epsilon_0 = 8.85 \cdot 10^{-12} \frac{\text{C}^2}{\text{N}\cdot\text{m}^2}$ is the permittivity of free space.

If we have several point charges, at different distances from Q, we have an electric field:

$$\mathbf{F} = Q\mathbf{E}, \quad (2.3)$$

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i^2} \hat{\mathbf{r}}_i. \quad (2.4)$$

If we have a continuous charge distribution, we can calculate the electric field by:

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \frac{1}{r^2} \hat{\mathbf{r}} dq, \quad (2.5)$$

where dq is the charge in a small element, be it line charge ($dq = \lambda dl$), area charge ($dq = \sigma da$) or volume charge ($dq = \rho d\tau$).

Special cases:

Field of infinite length wire:

$$E = \frac{1}{4\pi\epsilon_0} \frac{2\lambda}{z}$$

Field of infinite plane:

$$\mathbf{E} = \frac{\sigma}{2\epsilon_0} \hat{\mathbf{a}}$$

Field outside a uniformly charged sphere with total charge q , $r > R$:

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$$

2.2 Divergence and Curl

Field lines are imaginary lines used to represent the direction and strength of a field, such as an electric field. They start on positive charges and end on negative charges. The density of these lines indicates the field's strength; closer lines mean a stronger field.

Flux refers to the quantity of a field passing through a given surface. For an electric field, it is the product of the electric field strength and the area through which the field lines pass, considering the angle between the field lines and the surface.

$$\Phi = \int_S \mathbf{E} \cdot d\mathbf{a} \quad (2.11)$$

Gauss's law states that the total electric flux through a closed surface is proportional to the charge enclosed within that surface.

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0}, \quad (2.13)$$

where Q_{enc} is the charge enclosed by the surface. It can be rewritten with the divergence theorem, when $Q_{\text{enc}} = \int_V \rho d\tau$:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}. \quad (2.14)$$

To calculate the curl of the electric field we see that:

$$\int_a^b \mathbf{E} \cdot d\mathbf{l} = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r_a} - \frac{q}{r_b} \right), \quad (2.18)$$

which on a closed interval gives us:

$$\oint \mathbf{E} \cdot d\mathbf{l} = 0, \quad (2.19)$$

$$\nabla \times \mathbf{E} = 0. \quad (2.20)$$

2.3 Electric potential

Since we have $\oint \mathbf{E} \cdot d\mathbf{l}$, it must be independent of the path taken, and therefore we can calculate the potential of a point as:

$$V(\mathbf{r}) = - \int_{\mathcal{O}}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l}, \quad (2.21)$$

and the electric potential difference between two points as:

$$V(\mathbf{b}) - V(\mathbf{a}) = - \int_a^b \mathbf{E} \cdot d\mathbf{l}, \quad (2.22)$$

$$\Rightarrow \mathbf{E} = -\nabla V. \quad (2.23)$$

Poisson's equation:

$$\nabla^2 V = -\frac{\rho}{\epsilon_0}, \quad (2.24)$$

which gives us a way to calculate the potential from the charge distribution:

For a collection of point charges:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i}, \quad (2.27)$$

For a continuous charge distribution:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{1}{r} dq, \quad (2.28)$$

For a volume charge distribution:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{r} d\tau'. \quad (2.29)$$

Special cases:

Potential of infinite length wire:

$$V = \frac{1}{4\pi\epsilon_0} \lambda \ln\left(\frac{b}{a}\right)$$

Potential of infinite plane:

$$V = \frac{\sigma}{2\epsilon_0} z$$

Potential of spherical shell:

$$V = \begin{cases} \frac{1}{4\pi\epsilon_0} \frac{q}{r} & r > R \\ \frac{1}{4\pi\epsilon_0} \frac{q}{R} & r < R \end{cases}$$

Figure (2.35) shows the way to go between electric field, electric potential and volume charge:

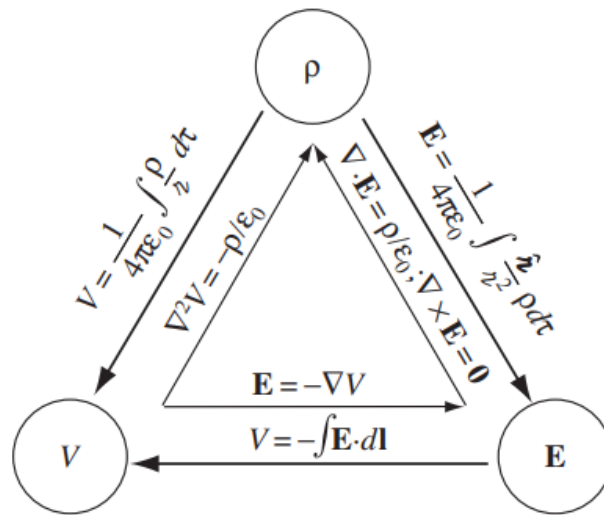


FIGURE 2.35

2.4 Work and Energy

If you want to move a test charge (Q) from point $\mathbf{a}$ to point $\mathbf{b}$, you have to do work against the electric field. The work done is the potential difference between the two points:

$$W = Q [V(\mathbf{b}) - V(\mathbf{a})]. \quad (2.38)$$

So if you want to move the point from infinity to a point $\mathbf{r}$, and you have the reference point in infinity:

$$W = QV(\mathbf{r}). \quad (2.39)$$

The work to move a collection of charges is:

$$W = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \sum_{j>i}^n \frac{q_i q_j}{r_{ij}}, \quad (2.40)$$

$$= \frac{1}{8\pi\epsilon_0} \sum_{i=1}^n \sum_{j \neq i}^n \frac{q_i q_j}{r_{ij}}, \quad (2.41)$$

$$= \frac{1}{2} \sum_{i=1}^n q_i V(\mathbf{r}_i). \quad (2.42)$$

For continuous charge distributions:

$$W = \frac{1}{2} \int V dq, \quad (2.43)$$

where again, dq is the charge in a small element, be it line charge ($dq = \lambda dl$), area charge ($dq = \sigma da$) or volume charge ($dq = \rho d\tau$).

If we take (2.43) for volume over all space, we get:

$$W = \frac{\epsilon_0}{2} \int E^2 d\tau. \quad (2.45)$$

Special cases

Energy of Q spread uniformly throughout sphere of radius R :

$$\begin{aligned} W_{tot} &= W_{out} + W_{in} \\ &= \frac{Q^2}{8\pi\epsilon_0 R} + \frac{3Q^2}{20\pi\epsilon_0 R} \end{aligned}$$

2.5 Conductors

A conductor is a material where the electrons are free to move around. The electric field inside a conductor is zero, and the charges are distributed on the surface, like metal. The electric field inside a conductor is zero, and when it is placed in an electric field, the charges collect on the surface to create an electrical field inside the conductor, that cancels out the initial field.

$$\nabla \mathbf{E} = \frac{\rho}{\epsilon_0} = 0$$

Induced charges on a conductor are always on the surface, and the electric field is always perpendicular to the surface. For a conductor with a cavity, the electric field is zero inside the cavity, and the charges are on the surface of the cavity. Outside the conductor, the electric field is the same as if the conductor was not there.

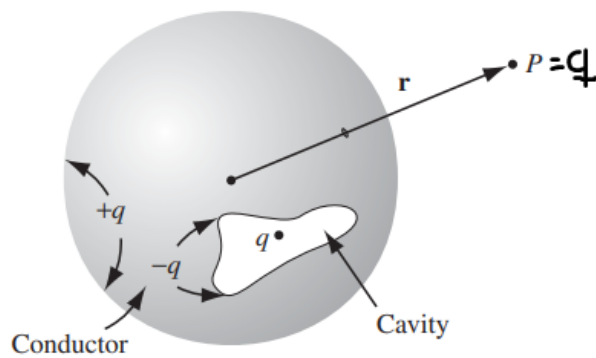


FIGURE 2.46

Figure 3.24 shows a conductor in an electric field, and how it would bend the field lines, so they are always perpendicular. so the electric field parallel to a conductor is 0.

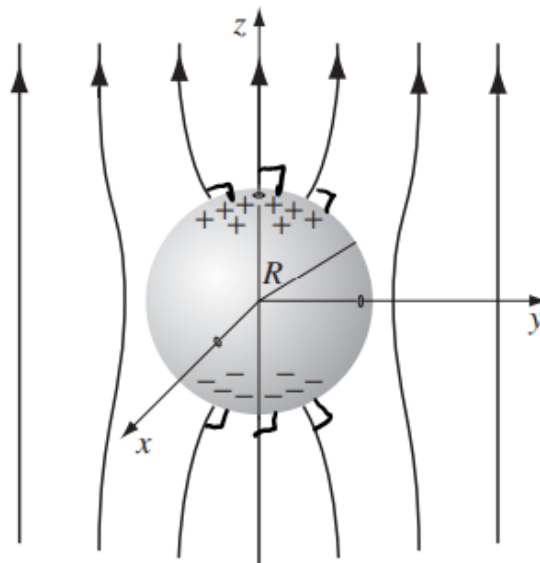


FIGURE 3.24

Because the field inside a conductor is zero, the field immediately outside is:

$$\mathbf{E} = \frac{\sigma}{\epsilon_0} \hat{\mathbf{n}}, \quad (2.48)$$

The force per unit area and outward electrostatic pressure on the surface

of a conductor is:

$$\mathbf{f} = \frac{1}{2\epsilon_0} \sigma^2 \hat{\mathbf{n}}, \quad (2.51)$$

$$P = \frac{\epsilon_0}{2} E^2 = \frac{\sigma^2}{2\epsilon_0}. \quad (2.52)$$

Capacitors

If we have two conductors with charge Q and $-Q$, the electric potential between them is:

$$V = V_+ - V_- = - \int_{(-)}^{(+)} \mathbf{E} \cdot d\mathbf{l},$$

if the shape is complicated, $\mathbf{E}$ is hard to calculate, but we know it is proportional to Q , and therefore V is also proportional, which gives us the capacitance:

$$C = \frac{Q}{V}. \quad (2.53)$$

We can talk about the capacitance of a single capacitor, this refers to (2.53) where the reference point for V is infinity.

For a parallel-plate capacitor the capacitance is:

$$C = \frac{A\epsilon_0}{d}, \quad (2.54)$$

To charge a capacitor you have to move the charge from the positive plate to the negative plate. the energy required to do this is:

$$W = \frac{1}{2} CV^2 = \frac{1}{2} QV = \frac{1}{2} \frac{Q^2}{C}. \quad (2.55)$$

3 Potential

3.1 Laplace's equation

To easily calculate the potential in a region, we can use Laplace's equation:

$$\nabla^2 V = 0. \quad (3.4)$$

In one dimension the solution is:

$$V(x) = ax + b, \quad (3.6)$$

but for 2 dimensions it is not easy to calculate exactly, but you can calculate $V(x, y)$ as the average of the potential around the point:

$$V(x, y) = \frac{1}{2\pi R} \oint_{circle} V dl,$$

which might make no sense, but it can be used to calculate the charge distribution, when boundary conditions are known.

In three dimensions the value of V at point $\mathbf{r}$ is the average value of V over a spherical surface of radius R centered at $\mathbf{r}$:

$$V(\mathbf{r}) = \frac{1}{4\pi R^2} \oint_{\text{sphere}} V da.$$

First uniqueness theorem: The solution to Laplace's equation in some volume $\mathcal{V}$ is uniquely determined if V is specified on the boundary surface $\mathcal{S}$. Meaning that it doesn't matter how you get your solution, if it satisfies Laplace's equation, and the boundary values are correct, then the answer is correct.

Second uniqueness theorem: In a volume $\mathcal{V}$ surrounded by conductors and containing a specified charge density ρ , the electric field is uniquely determined if the *total charge* on each conductor is given. (The region as a whole can be bounded by another conductor, or else unbounded.)

3.2 The Method of Images

The method of images can be used to solve advanced electrostatic problems, by replacing conductors with fictitious charges that mimic the same boundary conditions.

For example, we have an infinite grounded plate, with a point charge a distance d above it. The boundary potential is that $V = 0$ when $z = 0$ and that $V \rightarrow 0$ far from the charge (Fig 3.10). The same boundary conditions are satisfied if we have a point charge at $-d$ instead of the conductor (Fig 3.11). Therefore the potential for a pointcharge above an infinite grounded plate is:

$$V(x, y, z) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{\sqrt{x^2 + y^2 + (z - d)^2}} - \frac{q}{\sqrt{x^2 + y^2 + (z + d)^2}} \right]. \quad (3.9)$$

From this we can calculate the induced surface charge on the conductor:

$$\begin{aligned} \sigma(x, y) &= -\epsilon_0 \frac{\partial V}{\partial z} \Big|_{z=0} \\ &= \frac{-qd}{2\pi(x^2 + y^2 + d^2)^{3/2}}, \end{aligned} \quad (3.10)$$

and the total charge:

$$Q = \int \sigma da = -q. \quad (3.11)$$

We can also use the method of images to find the force on the point charge due to the induction:

$$\mathbf{F} = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{(2d)^2} \hat{\mathbf{z}}, \quad (3.12)$$

but the work will not be the same for the plane, and the point charge, so if we want to calculate the energy to move the charge from infinity to d as:

$$W = \int_{\infty}^d \mathbf{F} \cdot d\mathbf{l} = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{4d}. \quad (3.14)$$

You can not put the image charge in the region where you are calculating the potential, so in example 3.2, where we have a point charge a distance a away from a spherical grounded conductor, the image charge is inside the sphere.

This gives the potential:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{\tau} + \frac{q'}{\tau'} \right), \quad (3.17)$$

where q' is the image charge, and τ and τ' is the distance from q and q' .

3.3 Seperation of variables

Chapter 3.3.1 shows and 3 examples of how to calculate the potential between 2 parallel grounded conductors, connected by insulators. To calculate the potential between the plates we use separation of variables, so we assume the potential can be written as:

$$V(x, y) = X(x)Y(y) \quad (3.22)$$

$$\Rightarrow \frac{1}{X} \frac{\partial^2 X}{\partial x^2} + \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} = 0 \quad (3.23)$$

which we can then solve from the boundary conditions of the problem, with the help of Fouriers trick.

In chapter 3.3.2 we find a genrerall soultion for the potential in sperical coordinates, with symmetry in ϕ :

$$V(r, \theta) = \sum_{l=0}^{\infty} \left(A_l r^l + \frac{B_l}{r^{l+1}} \right) P_l(\cos(\theta)), \quad (3.65)$$

where A_l and B_l are constants determined by the boundary conditions, and $P_l(\cos(\theta))$ is the Legendre polynomials given by:

$$P_l(x) = \frac{1}{2^l l!} \left(\frac{d}{dx} \right)^l (x^2 - 1)^l. \quad (3.62)$$

$P_0(x) = 1$
$P_1(x) = x$
$P_2(x) = (3x^2 - 1)/2$
$P_3(x) = (5x^3 - 3x)/2$
$P_4(x) = (35x^4 - 30x^2 + 3)/8$
$P_5(x) = (63x^5 - 70x^3 + 15x)/8$

TABLE 3.1 Legendre Polynomials.

3.4 Multipole Expansion

A dipole is two opposite charges particles, collect two dipoles to make an quadrupole, and so on.

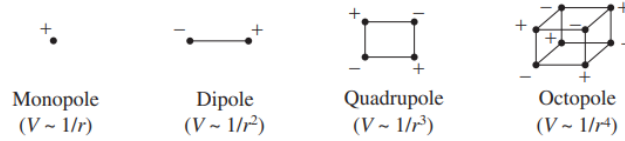


Fig. 3.27

To calculate the potential from a multipole we use:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \int (r')^n P_n(\cos(\alpha)) \rho(\mathbf{r}') d\tau', \quad (3.95)$$

where P_l is the Legendre polynomial, ρ is the total charge of the distribution. $n = 0$ for monopole, 1 for dipole and so on.

For monopole and dipole we get:

$$V_{mon}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}, \quad (3.97)$$

$$V_{dip}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^2}, \quad (3.99)$$

where $\mathbf{p}$ is the dipole moment of the distribution:

$$\mathbf{p} = \int \mathbf{r}' \rho(\mathbf{r}') d\tau'. \quad (3.98)$$

The electric field of a dipole is:

$$\mathbf{E}_{dip}(r, \theta) = \frac{p}{4\pi\epsilon_0 r^3} (2 \cos(\theta) \hat{\mathbf{r}} + \sin(\theta) \hat{\boldsymbol{\theta}}). \quad (3.103)$$

4 Electric Fields in Matter

4.1 Dielectrics

If you put a neutrol atom in an electric field, it is either ripped apart, being ionized, or it reaches an equilibrium, where the force from the electric field and the force holding the atom together cancel eachother, making the atom polarized. This will cause the atom to have a dipolemoment $\mathbf{p}$, with the same direction as $\mathbf{E}$.

$$\mathbf{p} = \alpha \mathbf{E}, \quad (4.1)$$

here α is called the atomic polarizability, and is dependent on the atoms structure.

If an atom was made by a center charge q surrounded by as spherical charged cloud $-q$ with radius a , then the electric field we place the atom in, would move the nucleus a distance d from the center. Now the electric field in equilibrium would be:

$$\begin{aligned} E &= \frac{1}{4\pi\epsilon_0} \frac{qd}{a^3}, \quad \text{or} \quad p = qd = (4\pi\epsilon_0 a^3) E, \\ \Rightarrow \alpha &= 4\pi\epsilon_0 a^3 = 3\epsilon_0 v, \end{aligned} \quad (4.2)$$

where v is the volume of the atom.

Table 4.1. Atomic polarizabilities ($\alpha/4\pi\epsilon_0$, in units of 10^{-30} m^3). Data from *Handbook of Chemistry and Physics*, 91st ed., CRC Press, Boca Raton (2010).

H	He	Li	Be	C	Ne	Na	Ar	K	Cs
0.667	0.205	24.3	5.60	1.67	0.396	24.1	1.64	43.4	59.4

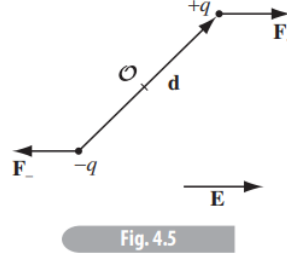
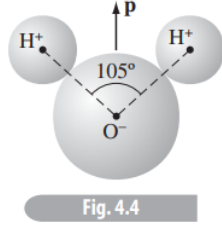
For molecules it is not so easy, as the dipolemoment might not be the same direction as $\mathbf{E}$, and therefore we get 9 polarizability constants:

$$\begin{cases} p_x = \alpha_{xx} E_x + \alpha_{xy} E_y + \alpha_{xz} E_z, \\ p_y = \alpha_{yx} E_x + \alpha_{yy} E_y + \alpha_{yz} E_z, \\ p_z = \alpha_{zx} E_x + \alpha_{zy} E_y + \alpha_{zz} E_z. \end{cases} \quad (4.3)$$

Alignment of polar molecules

Some molecules have their own dipolemoment, such as water, which are called polar molecules. If placed in an uniform electric field, the force on the postive end, cancels out with the force on the negative end, but there will be a torque

$$\mathbf{N} = (\mathbf{r}_+ \times \mathbf{F}_+) + (\mathbf{r}_- \times \mathbf{F}_-) = \mathbf{p} \times \mathbf{E}. \quad (4.4)$$



$\mathbf{N}$ is in such a direction to line $\mathbf{p}$ parallel to $\mathbf{E}$.

If the field is not uniform over det atom, which is rare, there is a net force on the atom

$$\mathbf{F} = (\mathbf{p} \cdot \nabla) \mathbf{E}. \quad (4.5)$$

An ideal dipole is the atrifical limit $d \rightarrow 0$ and $q \rightarrow \infty$. The energy of an ideal dipolemoment is

$$U = -\mathbf{p} \cdot \mathbf{E}, \quad (4.6)$$

and the interaction energy between two ideal dipoles with displacement $\mathbf{r}$ is

$$U = \frac{1}{4\pi\epsilon_0} \frac{1}{r^3} [\mathbf{p}_1 \cdot \mathbf{p}_2 - 3(\mathbf{p}_1 \cdot \hat{\mathbf{r}})(\mathbf{p}_2 \cdot \hat{\mathbf{r}})]. \quad (4.7)$$

If we place a material made of nonpolar molecules in an electric field, it will induce a dipole moment in all the molecules. If we place a polar material, the field will turn all the molecules. We get the polarization which is

$\mathbf{P}$ = dipole moment per unit volume.

4.2 Field of a Polarized Object

To calculate the field that arises from the induced dipoles in a material placed in an electric field, we first calculate the potential by integrating each dipole over det volume $\mathbf{p} = \mathbf{P}d\tau'$:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\mathbf{P}(\mathbf{r}') \cdot \hat{\mathbf{z}}}{z^2} d\tau', \quad (4.9)$$

$$= \frac{1}{4\pi\epsilon_0} \oint_S \frac{\sigma_b}{z} da' + \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho_b}{z} d\tau', \quad (4.13)$$

where we have the surface charge density $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$ and the volume charge density $\rho_b = -\nabla \cdot \mathbf{P}$.

For a uniformly polarized sphere of radius R , the volume charge density is zero baccuse $\mathbf{P}$ is uniform, but

$$\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}} = P \cos \theta,$$

from this we get

$$V(r, \theta) = \begin{cases} \frac{P}{3\epsilon_0} r \cos \theta & \text{for } r \leq R, \\ \frac{P}{3\epsilon_0} \frac{R^3}{r^2} \cos \theta & \text{for } r \geq R. \end{cases}$$

So the field inside the sphere is uniform, and outside the sphere it behaves like a ideal dipole, whose dipole moment is the total dipole moment of a sphere $\mathbf{p} = \frac{4}{3}\pi R^3 \mathbf{P}$.

4.3 Electric Displacement

Along with the bound charge, there is also free charge which could be electrons on a conductor, or ions embedded in the dielectric material, so any charge not associated with $\mathbf{P}$. Within the dielectric, the total charge density is

$$\rho = \rho_b + \rho_f, \quad (4.20)$$

and Gauss's law is

$$\epsilon_0 \nabla \cdot \mathbf{E} = \rho = \rho_b + \rho_f = -\nabla \cdot \mathbf{P} + \rho_f,$$

where $\mathbf{E}$ is the total field.

This leads to a term for ρ_f :

$$\begin{aligned} \rho_f &= \nabla \cdot (\epsilon_0 \mathbf{E} + \mathbf{P}), \\ &= \nabla \cdot \mathbf{D}, \end{aligned} \quad (4.22)$$

$$\Rightarrow \oint \mathbf{D} \cdot d\mathbf{a} = Q_{f_{enc}}. \quad (4.23)$$

Here $\mathbf{D}$ is known as the electric displacement.

Typically we know ρ_f , and we can therefore calculate $\mathbf{D}$. When computing the displacement look for symmetry, if there is spherical, cylindrical or plane symmetry we can use (4.23) because $\nabla \times \mathbf{D} = 0$.

4.4 Linear Dielectrics

The polarization is proportional to the electric field:

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}, \quad (4.30)$$

where χ_e is the electric susceptibility, which is dependant on the structure of the material. Materials that obey eq. 4.30 are called linear dielectrics. Notice that $\mathbf{E}$ is the total electric field, and therefore we can not calculate $\mathbf{P}$ from eq. 4.30, instead we could calculate

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} = \epsilon_0 (1 + \chi_e) \mathbf{E} = \epsilon \mathbf{E}, \quad (4.31)$$

where we have the permittivity of the material, and dielectric constant (Table 4.2):

$$\epsilon = \epsilon_0 (1 + \chi_e), \quad (4.33)$$

$$\epsilon_r = 1 + \chi_e = \frac{\epsilon}{\epsilon_0}. \quad (4.34)$$

When we have a charge distribution in homogeneous linear dielectric, the field is calculated as in vacuum, but using ϵ instead of ϵ_0 . To calculate the full

energy stored in a electrostatic system in a linear dielectric, we rewrite (2.45) to

$$W = \frac{\epsilon_0}{2} \int \epsilon_r E^2 d\tau = \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} d\tau. \quad (4.58)$$

Say we have a dielectric in a capacitor, the force required to remove the dielectric can be calculated as

$$F = -\frac{dW}{dx} = \frac{1}{2} V^2 \frac{dC}{dx}, \quad (4.64)$$

$$= -\frac{\epsilon_0 \chi_e w}{2d} V^2, \quad (4.65)$$

where d is the distance between the plates, and w is the width of the plate, perpendicular to the force.

5 Magnetostatics

5.1 Magnetic Forces

If we have a wire with a current running through, it will create a magnetic field, the direction of the field can be found by the right hand rule, if your thumb is in the direction of current, your fingers curl in the direction of the magnetic field.

A force will be induced on a moving charge (Q) in a magnetic field, moving with velocity ($\mathbf{v}$)

$$\mathbf{F}_{mag} = Q (\mathbf{v} \times \mathbf{B}). \quad (5.1)$$

This is the Lorentz force law. When there is both a magnetic field and an electric field, the force would be

$$\mathbf{F} = Q (\mathbf{E} + (\mathbf{v} \times \mathbf{B})). \quad (5.2)$$

For a charged particle in a magnetic field, the magnetic force will create a centripetal acceleration, which leads to the cyclotron formula, used to find the momentum of a charged particle,

$$QvB = m \frac{v^2}{R}, \quad \text{or } p = QBR. \quad (5.3)$$

If we have both a magnetic field and an electric field, we can follow example 5.2, to show that the particle will move in a cycloid motion, a circle where the center moves $(0, R\omega t, R)$:

$$(y - R\omega t)^2 + (z - R)^2 = R^2 \quad (5.9)$$

$$R\omega = \frac{E}{B}. \quad (5.10)$$

A magnetic field can not speed up or slow down a particle, it can only change its direction, therefore the field does no work on the particle,

$$dW_{mag} = \mathbf{F}_{mag} \cdot d\mathbf{l} = Q(\mathbf{v} \times \mathbf{B}) \cdot \mathbf{v} dt = 0. \quad (5.11)$$

Current

A line charge λ traveling down a wire at speed v , constitutes a charge

$$\mathbf{I} = \lambda \mathbf{v}. \quad (5.14)$$

The magnetic force on a segment of current-carrying wire is:

$$\mathbf{F}_{mag} = \int (\mathbf{I} \times \mathbf{B}) d\mathbf{l}, \quad (5.15)$$

if $\mathbf{I}$ and $d\mathbf{l}$ point in the same direction:

$$\mathbf{F}_{mag} = I \int (d\mathbf{l} \times \mathbf{B}). \quad (5.17)$$

When charge moves over a surface we describe it by the surface current density

$$\mathbf{K} = \frac{d\mathbf{I}}{dl_{\perp}} = \sigma \mathbf{v}. \quad (5.23)$$

And the magnetic force on the surface current is

$$\mathbf{F}_{mag} = \int (\mathbf{K} \times \mathbf{B}) d\mathbf{a}. \quad (5.24)$$

In a volume we have the volume current density:

$$\mathbf{J} = \frac{d\mathbf{I}}{da_{\perp}} = \rho \mathbf{v}, \quad (5.26)$$

$$\mathbf{F}_{mag} = \int (\mathbf{J} \times \mathbf{B}) d\tau. \quad (5.27)$$

The total charge crossing a surface can be calculated as

$$I = \int_S \mathbf{J} d\mathbf{a}_{\perp} = \int_S \mathbf{J} d\mathbf{a}. \quad (5.28)$$

From this we get the continuity equation

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}. \quad (5.29)$$

5.2 Biot-Savart Law

The magnetic field of a steady line current is given by Biot-Savart law:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{I} \times \hat{\mathbf{r}}}{r^2} dl',$$

where μ_0 is the permeability of free space.

The magnitude of the magnetic field caused by an infinite wire is:

$$B = \frac{\mu_0 I}{2\pi s}, \quad (5.38)$$

where the direction follows the right-hand rule. For a point a distance z above a circular loop of radius R , the field is:

$$B(z) = \frac{\mu_0 I}{2} \frac{R^2}{(R^2 + z^2)^{3/2}}. \quad (5.41)$$

For surface and volume charge the Biot-Savart law becomes:

$$\begin{aligned} \mathbf{B}(\mathbf{r}) &= \frac{\mu_0}{4\pi} \int \frac{\mathbf{K} \times \hat{\mathbf{r}}}{r^2} da', \\ \mathbf{B}(\mathbf{r}) &= \frac{\mu_0}{4\pi} \int \frac{\mathbf{J} \times \hat{\mathbf{r}}}{r^2} d\tau'. \end{aligned} \quad (5.42)$$

5.3 Divergence and Curl of $\mathbf{B}$

For any wire running through a chosen loop, the integral of $\mathbf{B}$ is:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}, \quad (5.44)$$

where I_{enc} is the total current enclosed by the loop. If the flow of current is volume current density, the enclosed current is:

$$I_{\text{enc}} = \int \mathbf{J} \cdot d\mathbf{a}. \quad (5.45)$$

Applying Stokes law, gives us the curl of $\mathbf{B}$ for infinite wires:

$$\begin{aligned} \int (\nabla \times \mathbf{B}) \cdot d\mathbf{a} &= \mu_0 \int \mathbf{J} \cdot d\mathbf{a}, \\ \Rightarrow \nabla \times \mathbf{B} &= \mu_0 \mathbf{J}. \end{aligned} \quad (5.46)$$

To find the general curl of $\mathbf{B}$, we use Biot Savart law:

$$\nabla \times \mathbf{B} = \frac{\mu_0}{4\pi} \int \mathbf{J} \times \frac{\hat{\mathbf{r}}}{r^2} d\tau', \quad (5.51)$$

$$= \mu_0 \mathbf{J}. \quad (5.56)$$

For the divergence of $\mathbf{B}$ we get:

$$\nabla \cdot \mathbf{B} = 0. \quad (5.50)$$

Ampère's Law

Equation (5.56) is called Ampère's law in differential form, if we integrate we get:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 \int \mathbf{J} \cdot d\mathbf{a} = \mu_0 I_{\text{enc}}. \quad (5.57)$$

This plays the role of Gauss law for magnetostatics.

Special cases:

Field of infinite sheet with in xy-plane with $\mathbf{K} = K\hat{\mathbf{x}}$:

$$\mathbf{B} = \begin{cases} +\frac{\mu_0}{2}K\hat{\mathbf{y}} & \text{for } z < 0, \\ -\frac{\mu_0}{2}K\hat{\mathbf{y}} & \text{for } z > 0. \end{cases} \quad (5.58)$$

Field of long solenoid with n windings and radius R :

$$\mathbf{B} = \begin{cases} \mu_0 n I \hat{\mathbf{z}} & \text{for } r < R, \\ 0 & \text{for } r > R. \end{cases} \quad (5.59)$$

Field of a toroidal coil (Fig 5.37) with n turns a distance s form the symmetry axis:

$$\mathbf{B}(\mathbf{r}) = \begin{cases} \frac{\mu_0 n I}{2\pi s} \hat{\phi} & \text{inside coil,} \\ 0 & \text{outside coil.} \end{cases} \quad (5.60)$$

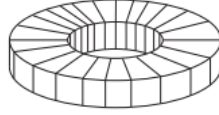


Fig. 5.37

Crosssection can be any shape

5.4 Magnetic Vector Potential

We introduce the vector potential $\mathbf{A}$:

$$\mathbf{B} = \nabla \times \mathbf{A}, \quad (5.61)$$

which divergence is 0:

$$\nabla \cdot \mathbf{A} = 0. \quad (5.63)$$

With the help of Amperes law this extends to:

$$\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}, \quad (5.64)$$

which gives us a direct term for $\mathbf{A}$, if $\mathbf{J}$ goes to zero at infinity:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}')}{r} d\tau'. \quad (5.65)$$

If $\mathbf{J}$ doesn't go to zero at infinity we can use:

$$\oint \mathbf{A} \cdot d\mathbf{l} = \int \mathbf{B} \cdot d\mathbf{a} = \phi. \quad (5.71)$$

Figure 5.51 shows the calculations between J, B and A.

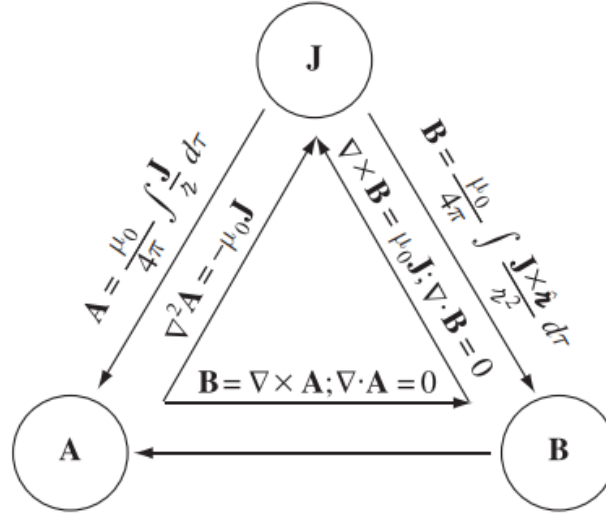


Fig. 5.51

If you want to find the vector potential for a current distribution, we can use multipole expansion:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \oint (r')^n P_n(\cos \alpha) d\mathbf{l}'. \quad (5.80)$$

which in most cases leads to the dipole moment being the dominant term:

$$\mathbf{A}_{\text{dip}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \hat{\mathbf{r}}}{r^2}, \quad (5.85)$$

where $\mathbf{m}$ is the magnetic dipole moment:

$$\mathbf{m} = I \int d\mathbf{a} = I\mathbf{a}. \quad (5.86)$$

The magnetic field of a magnetic dipole is:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi r^3} [3(\mathbf{m} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{m}]. \quad (5.89)$$

6 Magnetic Fields in Matter

6.1 Magnetization

Magnetic material, are magnetic because the electrons spin around their core. this creates magnetic dipoles. which in most magnets are turned in the direction of the magnetic field, the material is placed in.

If we put a rectangular loop in a uniform field, the loop experiences a torque:

$$\mathbf{N} = aF \sin \theta \hat{\mathbf{x}},$$

the magnitude of the force is $F = IbB$, which leads to:

$$\mathbf{N} = \mathbf{m} \times \mathbf{B}, \quad (6.1)$$

where $\mathbf{m} = Iab$ is the magnetic dipole moment of the loop.

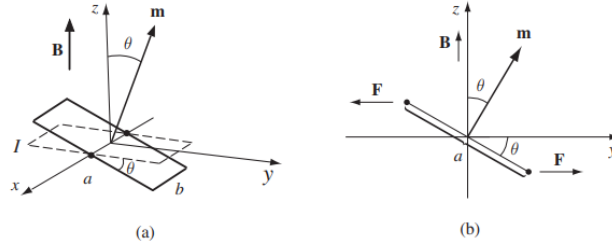


Fig. 6.2

In a uniform magnetic field the net force on a current loop is 0, but in a nonuniform field it is:

$$\mathbf{F} = \nabla(\mathbf{m} \cdot \mathbf{B}). \quad (6.3)$$

Atomic Orbits

If we say electrons revolve around the nucleus in a circular motion with radius R , it will look like a steady current:

$$I = \frac{e}{T} = \frac{e}{2\pi R/v},$$

$$\Rightarrow \mathbf{m} = -\frac{1}{2}evR\hat{\mathbf{z}}. \quad (6.4)$$

If we put an atom in a magnetic field, it will be hard to tilt the orbit of the electron, however the electron will speed up or slow down, depending on the direction of $\mathbf{B}$. Without the field we have:

$$\frac{1}{4\pi\epsilon_0} \frac{e^2}{R^2} = m_e \frac{v^2}{R}, \quad (6.5)$$

but in a field there is a force $-e(\mathbf{v} \times \mathbf{B})$, if $\mathbf{B}$ is perpendicular on the plane of orbit, we get:

$$\frac{1}{4\pi\epsilon_0} \frac{e^2}{R^2} + e\bar{v}B = m_e \frac{\bar{v}^2}{R}, \quad (6.6)$$

where $\bar{v}$ is the new speed of the electron. We can calculate the difference assuming the change in speed is small:

$$\Delta v = \bar{v} - v = \frac{eRB}{2m_e}. \quad (6.7)$$

This also leads to a change in dipole moment, opposite the field's direction:

$$\Delta \mathbf{m} = -\frac{1}{2}eR\Delta v\hat{\mathbf{z}} = -\frac{e^2R^2}{4m_e}\mathbf{B}. \quad (6.8)$$

Magnetization

So far we have seen two kinds of Magnetization. *Paramagnetism* where the dipoles associated with spins of unpaired electrons experience a torque tending to line them up parallel to the field, and *Diamagnetism* where the speed of the electrons is altered in such a way to change the orbital dipole moment in a direction opposite the field. These effects can be described by the *magnetization* $\mathbf{M}$, which is the magnetic dipole moment per unit volume.

6.2 Field of a Magnetized Object

With the magnetization we can calculate the vector potential from (5.85) knowing that $\mathbf{m} = \mathbf{M} d\tau'$:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{M}(\mathbf{r}') \times \hat{\mathbf{z}}}{r^2} d\tau', \quad (6.11)$$

which can also be rewritten in terms of the potential of a bound volume current and bound surface current,

$$\mathbf{J}_b = \nabla \times \mathbf{M}, \quad (6.13)$$

$$\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}. \quad (6.14)$$

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_V \frac{\mathbf{J}_b(\mathbf{r}')}{r} d\tau' + \frac{\mu_0}{4\pi} \oint_S \frac{\mathbf{K}_b(\mathbf{r}')}{r} da'. \quad (6.15)$$

6.3 Auxiliary Field

We can now collect the field due to bound charge, and the field due to the free charge.

$$\mathbf{J} = \mathbf{J}_b + \mathbf{J}_f. \quad (6.17)$$

With Ampere's law we get:

$$\begin{aligned} \frac{1}{\mu_0} (\nabla \times \mathbf{B}) &= \mathbf{J} = \mathbf{J}_b + \mathbf{J}_f = \mathbf{J}_f + (\nabla \times \mathbf{M}), \\ \Rightarrow \nabla \times \left(\frac{1}{\mu_0} \mathbf{B} - \mathbf{M} \right) &= \mathbf{J}_f, \end{aligned}$$

We define $\mathbf{H}$:

$$\mathbf{H} = \frac{1}{\mu_0} \mathbf{B} - \mathbf{M}, \quad (6.18)$$

$$\Rightarrow \nabla \times \mathbf{H} = \mathbf{J}_f, \quad (6.19)$$

and in integral form:

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{fenc}}, \quad (6.20)$$

where I_{fenc} is the free current through the loop.

6.4 Linear and Nonlinear Media

In paramagnetic and diamagnetic materials, the magnetization is sustained by the field. Which means $\mathbf{M}$ is proportional to $\mathbf{B}$ and therefore also $\mathbf{H}$:

$$\mathbf{M} = \chi_m \mathbf{H}, \quad (6.29)$$

where χ_m is the magnetic susceptibility, and is dependent of material. Materials that obey (6.29) are called linear media. This gives us:

$$\mathbf{B} = \mu \mathbf{H}, \quad (6.31)$$

$$\Rightarrow \mu = \mu_0(1 + \chi_m), \quad (6.32)$$

which is the permeability of the material (Table 6.1).

Ferromagnetism

In ferromagnetic materials, the alignment of dipoles is "frozen", because each dipole like to face the same way as their neighbours. We split the material up into domains, which can have different alignments, which explains why not all iron tools are magnetic. When a magnetic field is applied, the domains will compete and move their boundaries, so the domains with dipoles facing parallel to the field will grow.

7 Electrodynamics

7.1 Electromotive Force

To make current flow, you have to "push" the charges. For most materials the current density $\mathbf{J}$ is proportional to the force per unit charge $\mathbf{f}$:

$$\mathbf{J} = \sigma \mathbf{f}, \quad (7.1)$$

where σ is the conductivity of the material, which is sometimes written as resistivity $\rho = 1/\sigma$ (Table 7.1). Usually it is the electromagnetic forces that "pushes" and (7.1) becomes:

$$\mathbf{J} = \sigma(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \quad (7.2)$$

but the velocity of the charges is small enough that we can ignore the term, and we get Ohm's Law:

$$\mathbf{J} = \sigma \mathbf{E}, \quad (7.3)$$

which might not seem familiar, but if we use (7.3) to find the potential difference between two electrodes, we get:

$$V = IR. \quad (7.4)$$

As electrons travel down a wire, they collide with the atoms in the material, and each other. This results in energy given off as heat in the resistor:

$$P = VI = I^2 R. \quad (7.7)$$

The reason a light turns on instantly when you close a switch, is because if the charges built up at a point in a circuit, it will create an electric field, that will then restore uniform flow. This electric field propagates through the circuit at lightspeed, hence why the light turns on almost instantly.

So there are two forces that drive the current, the electric field and the source, often chemical force from a battery:

$$\mathbf{f} = \mathbf{f}_s + \mathbf{E}. \quad (7.8)$$

The electromotive force or *emf*, is determined by:

$$\mathcal{E} = \oint \mathbf{f} \cdot d\mathbf{l} = \oint \mathbf{f}_s \cdot d\mathbf{l}. \quad (7.9)$$

In an ideal emf we have $\mathbf{E} = -\mathbf{f}_s$, and therefore:

$$V = \int_a^b \mathbf{f}_s \cdot d\mathbf{l} = \mathcal{E}. \quad (7.10)$$

Motional emf

Suppose you move part of a current loop through a uniform magnetic field, with Lorentz law this will induce a current in the loop.

$$\mathcal{E} = \oint \mathbf{f}_{\text{mag}} \cdot d\mathbf{l} = vBh, \quad (7.11)$$

where the loop is rectangular with height h . But the magnetic force does not do work on the loop, so the person moving the loop is doing the work. The force required is $f_{\text{pull}} = uB$ where u is the vertical speed of the charges. This leads to the work done by unit charge:

$$\oint \mathbf{f}_{\text{pull}} \cdot d\mathbf{l} = vBh = \mathcal{E}.$$

The flux of $\mathbf{B}$ through a loop is:

$$\phi = \int \mathbf{B} \cdot d\mathbf{a}. \quad (7.12)$$

Instead of calculating for each charge, we use the flux through the loop:

$$\begin{aligned} \phi &= xBh, \quad \frac{d\phi}{dt} = -vBh, \\ \Rightarrow \mathcal{E} &= -\frac{d\phi}{dt}. \end{aligned} \quad (7.13)$$

7.2 Electromagnetic Induction

Faraday found that if he moved the magnetic field instead of the loop, current would still run through the loop. But it can't be the magnetic field moving the charges, instead it is an induced electric field.

$$\mathcal{E} = \oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\phi}{dt}, \quad (7.23)$$

this leads to Faraday's law:

$$\oint \mathbf{E} \cdot d\mathbf{l} = - \int \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{a} \quad (7.24)$$

$$\nabla \times \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t}. \quad (7.25)$$

Toghether with this Lenz's law say that the induced current is in such a direction that it opposes the change in magnetic flux.

If $\mathbf{E}$ is a pure Faraday field, due only to the change in $\mathbf{B}$, then the calculations are analogue to magnetostatics:

$$\mathbf{E} = - \frac{1}{4\pi} \frac{\partial}{\partial t} \int \frac{\mathbf{B} \times \hat{\mathbf{r}}}{r^2} d\tau, \quad (7.27)$$

and if we have symmetry we can use Amperes law:

$$\oint \mathbf{E} \cdot d\mathbf{l} = - \frac{d\phi}{dt}. \quad (7.28)$$

Inductance

If we have two loops of wire, and a current runs in one of the wires, this creates a magnetic field. This magnetic field will have a flux through the other loop, which is proportional to the current:

$$\phi = MI, \quad (7.31)$$

where M is the mutual inductance. M is a geometrical property, and is the same for if the current was running in loop 2.

$$M_{12} = M_{21} = \frac{\mu_0}{4\pi} \oint \oint \frac{d\mathbf{l}_2 \cdot d\mathbf{l}_1}{r}. \quad (7.32)$$

If the current is not steady we see that an emf is induced in the other loop:

$$\mathcal{E}_2 = - \frac{d\phi_2}{dt} = -M \frac{dI_1}{dt}. \quad (7.34)$$

On top of that it also induces an emf in itself:

$$\phi = LI, \quad (7.35)$$

where L is the self-inductance, or just inductance. If the current is changing we get:

$$\mathcal{E} = -L \frac{dI}{dt}. \quad (7.36)$$

When you close a circuit with an inductance L and resistance R , the current will build up, since

$$\begin{aligned} \mathcal{E}_0 - L \frac{dI}{dt} &= IR, \\ \Rightarrow I(t) &= \frac{\mathcal{E}_0}{R} (1 - e^{-(R/L)t}), \end{aligned} \quad (7.38)$$

if $I(0) = 0$.

It takes energy to build up this current in the loop, which can be seen as energy stored in the magnetic field. If we start at $I(0) = 0$ and work up to I , we get:

$$W = \frac{1}{2}LI^2, \quad (7.39)$$

$$= \frac{1}{2} \oint (\mathbf{A} \cdot \mathbf{I}) dl, \quad (7.40)$$

$$= \frac{1}{2} \int_{\mathcal{V}} (\mathbf{A} \cdot \mathbf{J}) d\tau, \quad (7.41)$$

$$= \frac{1}{2\mu_0} \int_{\text{all space}} B^2 d\tau. \quad (7.44)$$

7.3 Maxwell's Equations

So far we have 4 equations:

$$\begin{aligned} (i) \quad \nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0} && \text{(Gauss's law),} \\ (ii) \quad \nabla \cdot \mathbf{B} &= 0 && \text{(No name),} \\ (iii) \quad \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} && \text{(Faraday's law),} \\ (iv) \quad \nabla \times \mathbf{B} &= \mu_0 \mathbf{J} && \text{(Ampère's law).} \end{aligned}$$

But there is a problem with Ampère's law, since it is only for magnetostatics. Maxwell fixed this problem and the (iv) equation becomes:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (7.46)$$

Griffiths write the Maxwell equations like this:

$$\left. \begin{aligned} (i) \quad \nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0}, & (iii) \quad \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} &= \mathbf{0}, \\ (ii) \quad \nabla \cdot \mathbf{B} &= 0, & (iv) \quad \nabla \times \mathbf{B} - \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} &= \mu_0 \mathbf{J}. \end{aligned} \right\} \quad (7.52)$$

Maxwell's Equations in Matter

If the polarization of a material changes it induces a current:

$$\mathbf{J}_p = \frac{\partial \mathbf{P}}{\partial t}, \quad (7.58)$$

which means that the total current density is:

$$\mathbf{J} = \mathbf{J}_f + \mathbf{J}_b + \mathbf{J}_p = \mathbf{J}_f + \nabla \times \mathbf{M} + \frac{\partial \mathbf{P}}{\partial t}. \quad (7.60)$$

With this we can write Maxwell's equations for free charge and current:

$$\left. \begin{aligned} (i) \quad \nabla \cdot \mathbf{D} &= \rho_f, & (iii) \quad \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ (ii) \quad \nabla \cdot \mathbf{B} &= 0, & (iv) \quad \nabla \times \mathbf{H} &= \mathbf{J}_f + \frac{\partial \mathbf{H}}{\partial t}, \end{aligned} \right\} \quad (7.65)$$

where

$$\begin{aligned} \mathbf{D} &= \varepsilon \mathbf{E} \quad \text{and} \quad \mathbf{H} = \frac{1}{\mu} \mathbf{B}, \\ \varepsilon &= \varepsilon_0(1 + \chi_e) \quad \text{and} \quad \mu = \mu_0(1 + \chi_m). \end{aligned} \quad (7.67)$$

8 Conservation Laws

8.1 Charge and energy

Because of local charge conservation, charge must flow through a surface to leave a system:

$$\frac{dQ}{dt} = - \oint_S \mathbf{J} \cdot d\mathbf{a}, \quad (8.2)$$

which leads to the continuity equation:

$$\frac{d\rho}{dt} = -\nabla \cdot \mathbf{J}. \quad (8.4)$$

From the work required to assemble a static charge distribution (2.45), and the work required to get current to move against the back emf (7.44), we get the total stored energy in electromagnetic fields, per unit volume:

$$u = \frac{1}{2} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right). \quad (8.5)$$

The rate of work done on all charges in a volume is:

$$\frac{dW}{dt} = \int \mathbf{E} \cdot \mathbf{J} d\tau, \quad (8.6)$$

which can be rewritten using Ampère-Maxwell law:

$$\frac{dW}{dt} = -\frac{d}{dt} \int_V \frac{1}{2} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) d\tau - \frac{1}{\mu_0} \oint_S (\mathbf{E} \times \mathbf{B}) \cdot d\mathbf{a}. \quad (8.9)$$

This is Poynting's theorem. The first integral is the total energy stored in the electromagnetic field, the second integral is the energy leaving the volume V . The energy per unit time, per unit area, transported by the fields is called the Poynting vector:

$$\mathbf{S} = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}). \quad (8.10)$$

Now we can rewrite (8.9) again:

$$\frac{dW}{dt} = -\frac{d}{dt} \int_V u d\tau - \oint_S \mathbf{S} \cdot d\mathbf{a}. \quad (8.11)$$

This leads to a general continuity equation for energy in electromagnetic fields:

$$\frac{\partial u}{\partial t} = -\nabla \cdot \mathbf{S}. \quad (8.12)$$

8.2 Momentum

The force per unit volume on charges is:

$$\mathbf{f} = \rho \mathbf{E} + \mathbf{J} \times \mathbf{B}, \quad (8.14)$$

$$\begin{aligned} &= \epsilon_0 [(\nabla \cdot \mathbf{E})\mathbf{E} + (\mathbf{E} \cdot \nabla)\mathbf{E}] + \frac{1}{\mu_0} [(\nabla \cdot \mathbf{B})\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{B}] \\ &\quad - \frac{1}{2} \nabla \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) - \epsilon_0 \frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{B}). \end{aligned} \quad (8.16)$$

which can be rewritten using Maxwell stress tensor:

$$T_{ij} = \epsilon_0 \left(E_i E_j - \frac{1}{2} \delta_{ij} E^2 \right) + \frac{1}{\mu_0} \left(B_i B_j - \frac{1}{2} \delta_{ij} B^2 \right). \quad (8.17)$$

This leads to force per unit volume:

$$\mathbf{f} = \nabla \cdot \mathbf{T} - \epsilon_0 \mu_0 \frac{\partial \mathbf{S}}{\partial t}, \quad (8.19)$$

and total force:

$$\mathbf{F} = \oint_S \mathbf{T} \cdot d\mathbf{a} - \epsilon_0 \mu_0 \frac{d}{dt} \int_V \mathbf{S} d\tau. \quad (8.20)$$

with this we can rewrite the force per unit volume and total force:

$$\mathbf{f} = \nabla \cdot \mathbf{T} - \epsilon_0 \mu_0 \frac{\partial \mathbf{S}}{\partial t}, \quad (8.19)$$

$$\mathbf{F} = \frac{d\mathbf{p}}{dt} = \oint_S \mathbf{T} \cdot d\mathbf{a} - \epsilon_0 \mu_0 \frac{d}{dt} \int_V \mathbf{S} d\tau. \quad (8.20)$$

this gives us an expression for the momentum of the charges stored in the field:

$$\mathbf{p} = \epsilon_0 \mu_0 \int_V \mathbf{S} d\tau, \quad (8.28)$$

the momentum density is:

$$\mathbf{g} = \epsilon_0 \mu_0 \mathbf{S} = \epsilon_0 (\mathbf{E} \times \mathbf{B}). \quad (8.29)$$

There can also be an angular momentum stored in the field, the angular momentum per unit volume is:

$$\mathbf{l} = \mathbf{r} \times \mathbf{g} = \epsilon_0 [\mathbf{r} \times (\mathbf{E} \times \mathbf{B})]. \quad (8.33)$$

8.3 Magnetic Forces Do No Work

According to the Lorentz force law,

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}),$$

the magnetic part of the force, $\mathbf{F}_{\text{mag}} = q \mathbf{v} \times \mathbf{B}$, is always perpendicular to the velocity $\mathbf{v}$ and therefore does no work:

$$\mathbf{F}_{\text{mag}} \cdot \mathbf{v} = 0.$$

This implies that magnetic fields cannot increase or decrease the kinetic energy of charged particles. Despite this, real-life systems (e.g., a magnetic crane lifting a car) seem to contradict this rule. However, Griffiths explains:

- The energy comes from external sources (e.g., power supplies), not from the magnetic field itself.
- Magnetic fields redirect forces but cannot do work.
- Induced electric fields (via Faraday's law) or motional emf are responsible for actual energy transfer.
- Examples (e.g., lifting current loops or magnets) show that the magnetic force is conservative, redistributing energy but not generating it.

Any scenario that appears to violate this must be explained through alternative mechanisms (e.g., induced electric fields, energy stored in fields, or mechanical agents).

with this we can rewrite the force per unit volume and total force:

$$\mathbf{f} = \nabla \cdot \mathbf{T} - \epsilon_0 \frac{\partial \mathbf{S}}{\partial t}, \quad (8.19)$$

$$\mathbf{F} = \frac{d\mathbf{p}}{dt} = \oint_S \mathbf{T} \cdot d\mathbf{a} - \epsilon_0 \mu_0 \frac{d}{dt} \int_V \mathbf{S} d\tau. \quad (8.20)$$

this gives us an expression for the momentum of the charges stored in the field:

$$\mathbf{p} = \epsilon_0 \mu_0 \int_V \mathbf{S} d\tau, \quad (8.28)$$

the momentum density is:

$$\mathbf{g} = \epsilon_0 \mu_0 \mathbf{S} = \epsilon_0 (\mathbf{E} \times \mathbf{B}). \quad (8.29)$$

There can also be an angular momentum stored in the field, the angular momentum per unit volume is:

$$\mathbf{l} = \mathbf{r} \times \mathbf{g} = \epsilon_0 [\mathbf{r} \times (\mathbf{E} \times \mathbf{B})]. \quad (8.33)$$

9 Electromagnetic Waves

9.1 Waves in One Dimension

Waves propagating in z with a speed v , satisfy the wave equation:

$$\frac{\partial^2 f}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 f}{\partial t^2}, \quad (9.2)$$

$v = T/\mu$ where T is the tension and μ is the mass per unit length. The wave equation is linear, so the sum of two solutions is a solution. We have 3 different kinds of wave equations:

Sinusoidal waves:

$$f(z, t) = A \cos [k(z - vt) + \delta], \quad (9.7)$$

where A is the amplitude, the cosine is the phase and δ is the phase constant. k is the wave number related to the wavelength λ :

$$\lambda = \frac{2\pi}{k}. \quad (9.8)$$

We can calculate the period T , the frequency ν and the angular frequency ω :

$$T = \frac{2\pi}{kv}, \quad (9.9)$$

$$\nu = \frac{1}{T} = \frac{kv}{2\pi}, \quad (9.10)$$

$$\omega = 2\pi\nu = kv. \quad (9.11)$$

The wavefunction can now be written as:

$$f(z, t) = A \cos(kz - \omega t + \delta). \quad (9.12)$$

Complex notation:

$$f(z, t) = \text{Re} [A e^{i(kz - \omega t + \delta)}] = \text{Re} [\tilde{f}(z, t)]. \quad (9.16)$$

Linear combination:

$$\tilde{f}(z, t) = \int_{-\infty}^{\infty} \tilde{A}_k e^{i(kz - \omega t + \delta)} dk. \quad (9.20)$$

If a wave hits a knot, that is where the tension on one side is different from the other, the wave equation becomes:

$$\tilde{f}(z, t) = \begin{cases} \tilde{A}_I e^{i(k_1 z - \omega t)} + \tilde{A}_R e^{i(-k_1 z - \omega t)} & \text{for } z < 0, \\ \tilde{A}_T e^{i(k_2 z - \omega t)} & \text{for } z > 0. \end{cases} \quad (9.25)$$

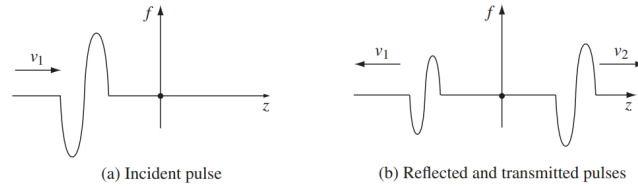


Fig. 9.5

We can find the amplitudes of the reflected and transmitted, from the incoming wave:

$$\tilde{A}_R = \frac{v_2 - v_1}{v_2 + v_1} \tilde{A}_I, \quad \tilde{A}_T = \frac{2v_1}{v_2 + v_1} \tilde{A}_I. \quad (9.30)$$

The wave can be polarized in the xy-plane with the vector:

$$\hat{n} = \cos \theta \hat{x} + \sin \theta \hat{y}. \quad (9.38)$$

9.2 Electromagnetic Waves in Vacuum

From Maxwell's equations (7.52, in vacuum) we can find that:

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}, \quad \nabla^2 \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}, \quad (9.41)$$

since $\nabla \cdot \mathbf{E} = 0$ and $\nabla \cdot \mathbf{B} = 0$. Since these are wave equations, we find that:

$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = c. \quad (9.43)$$

Electromagnetic waves are transverse waves, the magnetic and electric parts are perpendicular, therefore we get the relation:

$$\tilde{\mathbf{B}}_0 = \frac{k}{\omega} \hat{\mathbf{k}} \times \tilde{\mathbf{E}}_0, \quad (9.47)$$

$$B_0 = \frac{k}{\omega} E_0 = \frac{1}{c} E_0. \quad (9.48)$$

We can generalize the equations for B and E :

$$\begin{cases} \tilde{\mathbf{E}}(\mathbf{r}, t) = \tilde{E}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \hat{\mathbf{n}}, \\ \tilde{\mathbf{B}}(\mathbf{r}, t) = \tilde{B}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} (\hat{\mathbf{k}} \times \hat{\mathbf{n}}), \end{cases} \quad (9.50)$$

where $\hat{\mathbf{k}}$ is the wave propagation vector, and $\hat{\mathbf{n}}$ is the polarization vector.

In a monochromatic wave the energy per unit volume is (energy density):

$$u = \frac{1}{2} \epsilon_0 E^2 + \frac{1}{2 \mu_0} B^2 = \epsilon_0 E_0^2, \quad (9.56)$$

and the pointing vector is (energy flux density):

$$\mathbf{S} = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) = c u \hat{\mathbf{k}}. \quad (9.58)$$

Electromagnetic waves also carry momentum, and the momentum density is:

$$\mathbf{g} = \frac{1}{c^2} \mathbf{S}, \quad (9.59)$$

$$= \frac{1}{c} u \hat{\mathbf{k}}. \quad (9.60)$$

The average of a cosine is 1/2, so we get:

$$\langle u \rangle = \frac{1}{2} \epsilon_0 E_0^2, \quad (9.61)$$

$$\langle \mathbf{S} \rangle = \frac{1}{2} \epsilon_0 c E_0^2 \hat{\mathbf{k}}, \quad (9.62)$$

$$\langle \mathbf{g} \rangle = \frac{1}{2} \epsilon_0 E_0^2 \hat{\mathbf{k}}. \quad (9.63)$$

The average power per unit area transported by the wave is called intensity:

$$I = \langle \mathbf{S} \rangle = \frac{1}{2} \epsilon_0 c E_0^2, \quad (9.64)$$

and the average force per unit area (radiation pressure) is:

$$P = \frac{1}{A} \frac{\Delta p}{\Delta t} = \frac{1}{2} \epsilon_0 E_0^2 = \frac{I}{c}. \quad (9.65)$$

9.3 Electromagnetic Waves in Matter

Inside a linear meteriel where there is no free charge or free current, Maxwell's equations become:

$$\left. \begin{array}{ll} (i) & \nabla \cdot \mathbf{E} = 0, \\ (ii) & \nabla \cdot \mathbf{B} = 0, \end{array} \right\} \quad \left. \begin{array}{ll} (iii) & \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \\ (iv) & \nabla \times \mathbf{B} = \mu\epsilon \frac{\partial \mathbf{E}}{\partial t}. \end{array} \right\} \quad (9.68)$$

Eleectromagnetic waves propegate through a linear meteriel with a speed:

$$v = \frac{1}{\sqrt{\mu\epsilon}} = \frac{c}{n}, \quad (9.69)$$

where

$$n = \sqrt{\frac{\mu\epsilon}{\mu_0\epsilon_0}} \approx \sqrt{\epsilon_r}, \quad (9.70)$$

is the index of refraction.

Reflection and Transmission

When a wave travels from one material to another, it has to satisfy these conditions:

$$\left. \begin{array}{ll} (i) & \epsilon_1 E_1^\perp = \epsilon_2 E_2^\perp, \\ (ii) & B_1^\perp = B_2^\perp, \end{array} \right\} \quad \left. \begin{array}{ll} (iii) & \mathbf{E}_1^\parallel = \mathbf{E}_2^\parallel, \\ (iv) & \frac{1}{\mu_1} \mathbf{B}_1^\parallel = \frac{1}{\mu_2} \mathbf{B}_2^\parallel. \end{array} \right\} \quad (9.75)$$

If we use that $\mu_1 = \mu_2 = \mu$ we can see that the intensity of the electric field, when it goes from one medium to another, is:

$$E_R = \left| \frac{v_2 - v_1}{v_2 + v_1} \right| E_I, \quad E_T = \left(\frac{2v_2}{v_2 + v_1} \right) E_I, \quad (9.85)$$

$$E_R = \left| \frac{n_1 - n_2}{n_2 + n_1} \right| E_I, \quad E_T = \left(\frac{2n_1}{n_2 + n_1} \right) E_I. \quad (9.86)$$

and the intensity of the wave is:

$$R = \frac{I_R}{I_I} = \left(\frac{E_R}{E_I} \right)^2 = \left(\frac{n_1 - n_2}{n_1 + n_2} \right)^2, \quad (9.87)$$

$$T = \frac{I_T}{I_I} = \frac{\epsilon_2 v_2}{\epsilon_1 v_1} \left(\frac{E_T}{E_I} \right)^2 = \frac{4n_1 n_2}{(n_1 + n_2)^2}. \quad (9.88)$$

Reflection and Transmission at an Angle

When a wave goes from one medium to another at an angle, it will have to satisfy:

$$k_I v_1 = k_R v_1 = k_T v_2, \quad (9.92)$$

$$\mathbf{k}_I \mathbf{r} = \mathbf{k}_R \mathbf{r} = \mathbf{k}_T \mathbf{r}, \quad (9.95)$$

if we say that $(k_I)_y = 0$ we see that the incident, reflected and transmitted wave vectors form a plane of incident. We also get that:

$$k_I \sin \theta_I = k_R \sin \theta_R = k_T \sin \theta_T, \quad (9.99)$$

where the angles are measured from the normal of the plane of incident. The angle of incident is equal to the angle of refraction:

$$\theta_I = \theta_R. \quad (9.100)$$

Now we get Snell's law:

$$\frac{\sin \theta_T}{\sin \theta_I} = \frac{n_1}{n_2}. \quad (9.101)$$

If the polarization of the incident wave is parallel to the plane of incidence, we get Fresnel's equations:

$$\tilde{E}_R = \left(\frac{\alpha - \beta}{\alpha + \beta} \right) \tilde{E}_I, \quad \tilde{E}_T = \left(\frac{2}{\alpha + \beta} \right) \tilde{E}_I, \quad (9.110)$$

where

$$\alpha = \frac{\cos \theta_T}{\cos \theta_I}, \quad \beta = \frac{\mu_1 n_2}{n_1 \mu_2}. \quad (9.107)$$

At the Brewster angle, the reflected wave disappears:

$$\sin^2 \theta_B = \frac{1 - \beta^2}{(n_1/n_2)^2 - \beta^2}, \quad (9.112)$$

$$\Rightarrow \tan \theta_B \approx \frac{n_2}{n_1}. \quad (9.113)$$

We get the intensities:

$$I_I = \frac{1}{2} \epsilon_1 v_1 E_I^2 \cos \theta_I, \quad (9.114)$$

$$I_R = \frac{1}{2} \epsilon_1 v_1 E_R^2 \cos \theta_R \quad \text{and} \quad I_T = \frac{1}{2} \epsilon_2 v_2 E_T^2 \cos \theta_T \quad (9.115)$$

and the reflection and transmission coefficients:

$$R = \frac{I_R}{I_I} = \left(\frac{\alpha - \beta}{\alpha + \beta} \right)^2, \quad (9.116)$$

$$T = \frac{I_T}{I_I} = \alpha \beta \left(\frac{2}{\alpha + \beta} \right). \quad (9.117)$$

9.4 Absorption and Dispersion

When we are in a material with charge density ρ_f and current density $\mathbf{J}_f$, we see that the current density is proportional to the electric field:

$$\mathbf{J}_f = \sigma \mathbf{E}, \quad (9.119)$$

this leads to Maxwells equations being:

$$\left. \begin{aligned} (i) \quad \nabla \cdot \mathbf{E} &= \frac{1}{\epsilon} \rho_f, & (iii) \quad \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ (ii) \quad \nabla \cdot \mathbf{B} &= 0, & (iv) \quad \nabla \times \mathbf{B} &= \mu \sigma \mathbf{E} + \mu \epsilon \frac{\partial \mathbf{E}}{\partial t}. \end{aligned} \right\} \quad (9.120)$$

The continuity equation for free charge becomes:

$$\nabla \cdot \mathbf{J}_f = -\frac{\partial \rho_f}{\partial t}, \quad (9.121)$$

which, together with Ohm's law and Gauss's law, gives:

$$\rho_f(t) = e^{-(\sigma/\epsilon)t} \rho_f(0). \quad (9.122)$$

This gives that the initial free charge disappears in a time $\tau = \epsilon/\sigma$. In normal problems this time is small and we will wait for the free charge to disappear, which means (9.120 (i)) becomes $\nabla \cdot \mathbf{E} = 0$.

Having this current density will lead to the wavefunctions becomming:

$$\tilde{\mathbf{E}}(z, t) = \tilde{E}_0 e^{i(\tilde{k}z - \omega t)}, \quad \tilde{\mathbf{B}}(z, t) = \tilde{B}_0 e^{i(\tilde{k}z - \omega t)}, \quad (9.125)$$

where the wave number $\tilde{k}$ is complex:

$$\tilde{k}^2 = \mu \epsilon \omega^2 + i \mu \sigma \omega, \quad (9.126)$$

$$\tilde{k} = k + i\kappa, \quad (9.127)$$

where

$$\left. \begin{aligned} k &= \omega \sqrt{\frac{\epsilon \mu}{2}} \left[\sqrt{1 + \left(\frac{\sigma}{\epsilon \omega}\right)^2} + 1 \right]^{1/2} \\ \kappa &= \omega \sqrt{\frac{\epsilon \mu}{2}} \left[\sqrt{1 + \left(\frac{\sigma}{\epsilon \omega}\right)^2} - 1 \right]^{1/2} \end{aligned} \right\} \quad (9.128)$$

The imaginary part of the wave number results in a decresing amlitude with increasing z . The distance it takes to reduce the amplitude by a factor of $1/e$ is called the skin depth, which is a measure of how far the wave penetrates into the conductor.

$$d = \frac{1}{\kappa}. \quad (9.130)$$

k determines the wavelength, propegation speed and index of refraction:

$$\lambda = \frac{2\pi}{k}, \quad v = \frac{\omega}{k}, \quad n = \frac{ck}{\omega}. \quad (9.131)$$

The elctric and magnetic fields are:

$$\left. \begin{aligned} \mathbf{E}(z, t) &= E_0 e^{-\kappa z} \cos(kz - \omega t + \delta_E) \hat{\mathbf{x}}, \\ \mathbf{B}(z, t) &= B_0 e^{-\kappa z} \cos(kz - \omega t + \delta_E + \phi) \hat{\mathbf{y}}. \end{aligned} \right\} \quad (9.140)$$

We see that the magnetic field lacks behind by a factor:

$$\phi = \tan^{-1} \left(\frac{\kappa}{k} \right). \quad (9.136)$$

When there is free charge (9.75) becomes:

$$\left. \begin{aligned} (i) \quad \epsilon_1 E_1^\perp - \epsilon_2 E_2^\perp &= \sigma_f, & (iii) \quad \mathbf{E}_1^\parallel &= \mathbf{E}_2^\parallel, \\ (ii) \quad B_1^\perp &= B_2^\perp, & (iv) \quad \frac{1}{\mu_1} \mathbf{B}_1^\parallel - \frac{1}{\mu_2} \mathbf{B}_2^\parallel &= \mathbf{K}_f \times \hat{\mathbf{n}}. \end{aligned} \right\} \quad (9.141)$$

where σ_f is free surface charge, $\mathbf{K}_f$ is the free surface current and $\hat{\mathbf{n}}$ is perpendicular to the surface.

When a wave enters a material where different frequencies move at different speeds, the material is said to be dispersive. Like a raindrop that disperses white light into a rainbow. The speed of the different frequencies is given by:

$$v = \frac{\omega}{k}, \quad (9.151)$$

whereas the packet moves with the group velocity is:

$$v_g = \frac{d\omega}{dk}. \quad (9.152)$$

If we see the electrons, in a material, as charges on a spring, it will have a binding force:

$$F_b = -k_s \text{pring} x = -m\omega_0^2 x, \quad (9.153)$$

a damping force:

$$F_{damp} = -m\gamma \frac{dx}{dt}, \quad (9.154)$$

and, in the presence of an electromagnetic wave with frequency ω , a driving force:

$$F_d = qE = qE_0 \cos(\omega t), \quad (9.155)$$

where q is the charge of the electron and E_0 is the amplitude of the electric field, at the electron. Solving the differential equation stemming from Newton's second law, gives us a term for the system:

$$\tilde{x}(t) = \tilde{x}_0 e^{-i\omega t}, \quad (9.158)$$

$$\tilde{x}_0 = \frac{q/m}{\omega_0^2 - \omega^2 - i\gamma\omega} E_0 \quad (9.169)$$

where the resulting dipole moment is the real part of:

$$\tilde{p}(t) = q\tilde{x}(t). \quad (9.160)$$

In general, differently situated electrons within a given molecule experience different natural frequencies and damping coefficients. Let's say there are f_j electrons with frequency ω_j and damping γ_j in each molecule. If there are N molecules per unit volume, the polarization $\mathbf{P}$ is given by the real part of

$$\tilde{\mathbf{P}} = \frac{Nq^2}{m} \left(\sum_j \frac{f_j}{\omega_j^2 - \omega^2 - i\gamma_j\omega} \right) \tilde{\mathbf{E}}. \quad (9.161)$$

Now $\mathbf{P}$ and $\mathbf{E}$ are no longer proportional, but the complex part is:

$$\tilde{P} = \epsilon_0 \tilde{\chi}_e \tilde{E}. \quad (9.162)$$

So now that we have the complex susceptibility, we can write the complex permittivity:

$$\tilde{\epsilon} = \epsilon_0(1 + \tilde{\chi}_e),$$

and the complex dielectric constant:

$$\tilde{\epsilon}_r = \frac{\tilde{\epsilon}}{\epsilon_0} = 1 + \frac{Nq^2}{m\epsilon_0} \sum_j \frac{f_j}{\omega_j^2 - \omega^2 - i\gamma_j\omega}. \quad (9.163)$$

In a dispersive medium the wave equation has solutions of the form:

$$\tilde{\mathbf{E}}(z, t) = \tilde{E}_0 e^{-\kappa z} e^{i(\tilde{k}z - \omega t)}, \quad (9.168)$$

with the complex wave number:

$$\tilde{k} = \sqrt{\tilde{\epsilon}\mu_0}\omega = k + i\kappa, \quad (9.166)$$

$$= \frac{\omega}{c} \sqrt{\tilde{\epsilon}_r} \approx \frac{\omega}{c} \left[1 + \frac{Nq^2}{2m\epsilon_0} \sum_j \frac{f_j}{\omega_j^2 - \omega^2 - i\gamma_j\omega} \right]. \quad (9.171)$$

The index of refraction is:

$$n = \frac{ck}{\omega}, \quad (9.170)$$

$$\approx 1 + \frac{Nq^2}{2m\epsilon_0} \sum_j \frac{f_j(\omega_j^2 - \omega^2)}{(\omega_j^2 - \omega^2)^2 + \gamma_j^2\omega^2}. \quad (9.172)$$

10 Potentials and Fields

When working with the general form of Faraday's law (iii), the curl of $\mathbf{E}$ is not zero, therfor $\mathbf{E} \neq -\nabla V$. But since we have

$$\mathbf{B} = \nabla \times \mathbf{A}, \quad (10.2)$$

we can write Faraday's law as

$$\nabla \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} \right) = 0,$$

which leads to:

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t}. \quad (10.3)$$

With this new potential we can change Ampère's and Gauss's law:

$$\nabla^2 V + \frac{\partial}{\partial t}(\nabla \cdot \mathbf{A}) = -\frac{\rho}{\epsilon_0}, \quad (10.4)$$

$$(\nabla^2 \mathbf{A} - \mu_0\epsilon_0 \frac{\partial^2 \mathbf{A}}{\partial t^2}) - \nabla(\nabla \cdot \mathbf{A} + \mu_0\epsilon_0 \frac{\partial V}{\partial t}) = -\mu_0 \mathbf{J}. \quad (10.5)$$

For these equations we can change $\mathbf{A}$ and V with a gauge transformation, which does not affect the physical quantities of $\mathbf{E}$ or $\mathbf{B}$,

$$\left. \begin{aligned} \mathbf{A}' &= \mathbf{A} + \nabla \lambda(\mathbf{r}, t), \\ V' &= V - \frac{\partial}{\partial t} \lambda(\mathbf{r}, t). \end{aligned} \right\} \quad (10.7)$$

There are different kinds of gauge methods. The Columb gauge uses:

$$\nabla \cdot \mathbf{A} = 0, \quad (10.8)$$

which leads (10.4) to become:

$$\nabla^2 V = -\frac{\rho}{\epsilon_0}. \quad (10.9)$$

This is Poisson's equation which we can solve by:

$$V(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}', t)}{z} d\tau', \quad (10.10)$$

setting $V = 0$ at infinity. For equation (10.5) we get:

$$\nabla^2 \mathbf{A} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{A}}{\partial t^2} = -\mu_0 \mathbf{J} + \mu_0 \epsilon_0 \nabla \frac{\partial V}{\partial t}. \quad (10.11)$$

But this is not easier to calculate, so we use the Lorenz gauge:

$$\nabla \cdot \mathbf{A} = -\mu_0 \epsilon_0 \frac{\partial V}{\partial t}. \quad (10.12)$$

With this we get the equations to be:

$$\nabla^2 \mathbf{A} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{A}}{\partial t^2} = -\mu_0 \mathbf{J}, \quad (10.13)$$

$$\nabla^2 V - \mu_0 \epsilon_0 \frac{\partial^2 V}{\partial t^2} = -\frac{\rho}{\epsilon_0}. \quad (10.14)$$

Here we introduce the d'Alembertian (or box) operator:

$$\square^2 = \nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}, \quad (10.15)$$

which leads to the equations to become:

$$\left. \begin{aligned} \square^2 V &= -\frac{\rho}{\epsilon_0}, \\ \square^2 \mathbf{A} &= -\mu_0 \mathbf{J}. \end{aligned} \right\} \quad (10.16)$$

No we can calculate the potentials and fields with Jefimenko's equations:

$$\begin{aligned} V(\mathbf{r}, t) &= \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}', t_r)}{z} d\tau', \\ \mathbf{A}(\mathbf{r}, t) &= \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}', t_r)}{z} d\tau', \end{aligned} \quad (10.26)$$

$$\mathbf{E}(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int \left[\frac{\rho(\mathbf{r}', t_r)}{z^2} \hat{\mathbf{z}} + \frac{\dot{\rho}(\mathbf{r}', t_r)}{cz} \hat{\mathbf{z}} + \frac{\dot{\mathbf{J}}(\mathbf{r}', t_r)}{c^2 z} \right] d\tau', \quad (10.36)$$

$$\mathbf{B}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \left[\frac{\mathbf{J}(\mathbf{r}', t_r)}{z^2} + \frac{\dot{\mathbf{J}}(\mathbf{r}', t_r)}{cz} \right] \times \hat{\mathbf{z}} d\tau'. \quad (10.38)$$

Where $t_r = t - \frac{z}{c}$ is the retarded time.

Liénard-Wiechert Potentials

Now we can calculate the Liénard-Wiechert potentials V and $\mathbf{A}$, for a point charge q moving on a specified trajectory $\mathbf{w}(t)$, which is the position of the charge. Because of special relativity, the charge will move before we observe it, therefore we have a retarded time:

$$|\mathbf{r} - \mathbf{w}(t_r)| = c(t - t_r), \quad (10.44)$$

where $\mathbf{r}$ is the observation point, and $\mathbf{z}$ is the vector from the retarded point to the observation point,

$$\mathbf{z} = \mathbf{r} - \mathbf{w}(t_r). \quad (10.45)$$

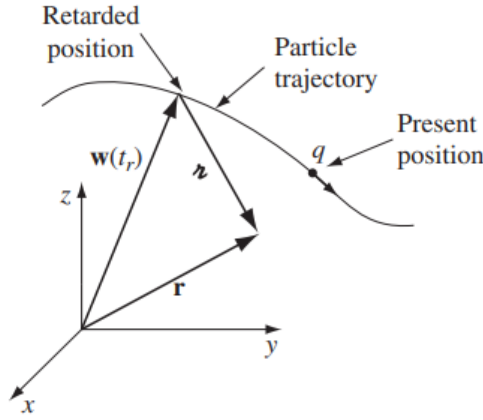


Fig. 10.8

This leads to the potential:

$$V(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \frac{qc}{(rc - \mathbf{z} \cdot \mathbf{v})}, \quad (10.46)$$

where $\mathbf{v}$ is the velocity of the charge at the retarded time. The vector potential is:

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \frac{qc\mathbf{v}}{(rc - \mathbf{z} \cdot \mathbf{v})} = \frac{\mathbf{v}}{c^2} V(\mathbf{r}, t). \quad (10.47)$$

Using these potentials we can calculate the fields:

$$\mathbf{E}(\mathbf{r}, t) = \frac{q}{4\pi\epsilon_0} \frac{z}{(\mathbf{z} \cdot \mathbf{u})^3} [(c^2 - v^2)\mathbf{u} + \mathbf{z} \times (\mathbf{u} \times \mathbf{a})], \quad (10.72)$$

$$\mathbf{B}(\mathbf{r}, t) = \frac{1}{c} \hat{\mathbf{z}} \times \mathbf{E}(\mathbf{r}, t) = \frac{q}{4\pi} \frac{\mathbf{u} \times \mathbf{a}}{(\mathbf{z} \cdot \mathbf{u})^3}, \quad (10.73)$$

where $\mathbf{u} = c\hat{\mathbf{z}} - \mathbf{v}$ and $\mathbf{a} = \frac{\partial \mathbf{v}}{\partial t}$ (acceleration of the particle at the retarded time).

11 Radiation

11.1 Dipole Radiation

When charges accelerate their fields transport energy out to infinity, we call it radiation. The power radiated from a sphere, of radius r , is the integral of the pointing vector, passing through the surface area.

$$P(r, t) = \oint \mathbf{S} \cdot d\mathbf{a}, \quad (11.1)$$

because of retardation the radiated power is:

$$P_{rad}(t_0) = \lim_{r \rightarrow \infty} P(r, t_0 + \frac{r}{c}). \quad (11.2)$$

Because of Jefimenko's equations, we know that only time-dependant fields that go like $1/r$ are responsible for the radiation.

For an electric dipole where the charge moves back and forth with angular frequency ω , we have the charge:

$$q(t) = q_0 \cos(\omega t), \quad (11.3)$$

and the dipole moment:

$$\mathbf{p}(t) = q_0 d \cos(\omega t) \hat{\mathbf{z}}, \quad (11.4)$$

where d is the distance between the two charges. From equation (10.26) we can see that the potential is:

$$V(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \left\{ \frac{q_0 \cos[\omega(t - z_+/c)]}{z_+} - \frac{q_0 \cos[\omega(t - z_-/c)]}{z_-} \right\}, \quad (11.5)$$

we can approximate this with the assumptions:

$$d \ll \frac{c}{\omega} \ll r,$$

$$\Rightarrow V(r, \theta, t) = -\frac{p_0 \omega}{4\pi\epsilon_0 c} \left(\frac{\cos \theta}{r} \right) \sin[\omega(t - r/c)], \quad (11.14)$$

and the vector potential can be approximated to:

$$\mathbf{A}(r, \theta, t) = -\frac{\mu_0 p_0 \omega}{4\pi r} \sin[\omega(t - r/c)] \hat{\mathbf{z}}. \quad (11.17)$$

From this we can also calculate the fields:

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t} = -\frac{\mu_0 p_0 \omega^2}{4\pi} \left(\frac{\sin \theta}{r} \right) \cos[\omega(t - r/c)] \hat{\boldsymbol{\theta}}, \quad (11.18)$$

$$\mathbf{B} = \nabla \times \mathbf{A} = -\frac{\mu_0 p_0 \omega^2}{4\pi c} \left(\frac{\sin \theta}{r} \right) \cos[\omega(t - r/c)] \hat{\boldsymbol{\phi}}. \quad (11.19)$$

The energy radiated by an oscilation is determined by the pointing vector:

$$\mathbf{S}(\mathbf{r}, t) = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) = \frac{\mu_0}{c} \left\{ \frac{p_0 \omega^2}{4\pi} \left(\frac{\sin \theta}{r} \right) \cos[\omega(t - r/c)] \right\}^2 \hat{\mathbf{r}}, \quad (11.20)$$

the intensity is:

$$\langle \mathbf{S} \rangle = \left(\frac{\mu_0 p_0^2 \omega^4}{32\pi^2 c} \right) \frac{\sin^2 \theta}{r^2} \hat{\mathbf{r}}, \quad (11.21)$$

notice that the intensity takes the form of a donut, as there is no radiation along the axis of the dipole. The total power radiated is:

$$\langle P \rangle = \int \langle \mathbf{S} \rangle \cdot d\mathbf{a} = \frac{\mu_0 p_0^2 \omega^4}{12\pi c}. \quad (11.22)$$

If we make a magnetic dipole, by sending a current through a wire loop, of radius b , we get the magnetic dipole moment:

$$\mathbf{m}(t) = \pi b^2 I(t) \hat{\mathbf{z}} = m_0 \cos(\omega t) \hat{\mathbf{z}}, \quad (11.24)$$

we get the vector potential, and fields, by making likewise approximations:

$$\mathbf{A}(r, \theta, t) = -\frac{\mu_0 m_0 \omega}{4\pi c} \left(\frac{\sin \theta}{r} \right) \sin[\omega(t - r/c)] \hat{\boldsymbol{\phi}}, \quad (11.35)$$

$$\mathbf{E}(r, \theta, t) = -\frac{\partial \mathbf{A}}{\partial t} = \frac{\mu_0 m_0 \omega^2}{4\pi c} \left(\frac{\sin \theta}{r} \right) \cos[\omega(t - r/c)] \hat{\boldsymbol{\phi}}, \quad (11.36)$$

$$\mathbf{B}(r, \theta, t) = \nabla \times \mathbf{A} = -\frac{\mu_0 m_0 \omega^2}{4\pi c^2} \left(\frac{\sin \theta}{r} \right) \cos[\omega(t - r/c)] \hat{\boldsymbol{\theta}}. \quad (11.37)$$

We see now that the intensity and power is:

$$\mathbf{S}(\mathbf{r}, t) = \frac{\mu_0}{c} \left\{ \frac{m_0 \omega^2}{4\pi c} \right\} \left(\frac{\sin \theta}{r} \right) \cos[\omega(t - r/c)]^2 \hat{\mathbf{r}}, \quad (11.38)$$

$$\langle \mathbf{S} \rangle = \left(\frac{\mu_0 m_0^2 \omega^4}{32\pi^2 c^3} \right) \frac{\sin^2 \theta}{r^2} \hat{\mathbf{r}}, \quad (11.39)$$

$$\langle P \rangle = \frac{\mu_0 m_0^2 \omega^4}{12\pi c^3}. \quad (11.40)$$

If we compare the two radiated powers from the electric and magnetic dipole, we see that the electric fields radiation is dominant.

For an electric dipole the power radiated can be approximated to:

$$P_{rad}(t_0) = \frac{\mu_0}{6\pi c} \left[\frac{\partial p(t_0)}{\partial t} \right]^2$$

11.2 Power Radiated by a Point Charge

Equation (10.72) and (10.73) gives us the fields of a charged particle in motion. The pointing vector is:

$$\mathbf{S} = \frac{1}{\mu_0 c} [E^2 \hat{\boldsymbol{\epsilon}} - (\hat{\boldsymbol{\epsilon}} \cdot \mathbf{E}) \mathbf{E}]. \quad (11.64)$$

Not all of the fields contribute to the radiation, some move with the particle. To calculate the radiated power, we integrate the pointing vector over a sphere of

radius τ , centered at the particle at time t . Since the surface area is proportional to τ^2 , only these parts contribute to the radiation.

$$\mathbf{E}_{rad} = \frac{q}{4\pi\epsilon_0} \frac{\tau}{(\boldsymbol{\tau} \cdot \mathbf{u})^3} [\boldsymbol{\tau} \times (\mathbf{u} \times \mathbf{a})], \quad (11.66)$$

since $\mathbf{E}_{rad}$ is perpendicular to $\hat{\boldsymbol{\tau}}$, we can write the pointing vector as:

$$\mathbf{S} = \frac{1}{\mu_0 c} E_{rad}^2 \hat{\boldsymbol{\tau}}. \quad (11.67)$$

If a particle is initially at rest, with instant acceleration $\mathbf{a}$, we can write the power radiated as:

$$P = \frac{\mu_0 q^2 a^2}{6\pi c}. \quad (11.70)$$

When the particle is moving with a velocity v , an effect like the doppler effect occurs. The radiated power then becomes:

$$P = \frac{\mu_0 q^2 \gamma^6}{6\pi c} \left(\mathbf{a}^2 - \left| \frac{\mathbf{v} \times \mathbf{a}}{c} \right|^2 \right), \quad (11.73)$$

where $\gamma = 1/\sqrt{1-v^2/c^2}$. This is the Liénard generalization of the Larmor formula (11.70). The γ^6 means that the radiation increases with velocity.

11.3 The Radiation Reaction

Because of energy conservation, when a particle radiates energy, the energy lost must correspond to an opposing force. But because of interaction of with non radiating fields, the energy lost is not strictly $\mathbf{F} \cdot \mathbf{v} = -P$. If we say that this is the average, we can find the radiation reaction force is:

$$\mathbf{F}_{rad} = \frac{\mu_0 q^2}{6\pi c} \dot{\mathbf{a}}, \quad (11.80)$$

this is the Abraham-Lorentz formula.

12 The Special Theory of Relativity

12.1 Relativistic Mechanics

From Special relativity we have time dialation which is noted by:

$$\Delta \bar{t} = \sqrt{1 - \frac{v^2}{c^2}} \Delta t, \quad (12.5)$$

which gives us the factor:

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}. \quad (12.6)$$

We also have length contraction:

$$\Delta \bar{x} = \gamma \Delta x. \quad (12.9)$$

With these two phenomenons we can see that the Lorentz transformation is:

$$\left. \begin{aligned} \bar{x} &= \gamma(x - vt), \\ \bar{y} &= y, \\ \bar{z} &= z, \\ \bar{t} &= \gamma\left(t - \frac{v}{c^2}x\right). \end{aligned} \right\} \quad (12.18)$$

And the other way around:

$$\left. \begin{aligned} x &= \gamma(\bar{x} + v\bar{t}), \\ y &= \bar{y}, \\ z &= \bar{z}, \\ t &= \gamma\left(\bar{t} + \frac{v}{c^2}\bar{x}\right). \end{aligned} \right\} \quad (12.19)$$

Four-Vector

This can be written on vector form by saying that $x^0 = ct$, $\beta = v/c$ and

$$x^1 = x, \quad x^2 = y, \quad x^3 = z, \quad (12.22)$$

and therefore we can write the Lorentz transformation as:

$$\begin{bmatrix} \bar{x}^0 \\ \bar{x}^1 \\ \bar{x}^2 \\ \bar{x}^3 \end{bmatrix} = \begin{bmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{bmatrix}. \quad (12.24)$$

Or in a single line:

$$\bar{x}^\mu = \Lambda^\mu_\nu x^\nu, \quad (12.25)$$

The dot product of two four-vectors is:

$$\mathbf{a} \cdot \mathbf{b} = a_\mu b^\mu = -a^0 b^0 + a^1 b^1 + a^2 b^2 + a^3 b^3, \quad (12.28)$$

where the subscript μ means covariant vector $a_\mu = (-a^0, a^1, a^2, a^3)$.

Special relativity also brings a relativistic momentum and energy:

$$\mathbf{p} = m\boldsymbol{\eta} = \gamma m\mathbf{v}, \quad (12.47)$$

where $\boldsymbol{\eta}$ is the relativistic velocity

$$\boldsymbol{\eta} = \gamma \mathbf{v}. \quad (12.40)$$

And the relativistic energy is:

$$E = \gamma mc^2. \quad (12.50)$$

Toghether we get the result:

$$E^2 - p^2 c^2 = m^2 c^4. \quad (12.55)$$

There is also a relativistic force,:

$$\bar{\mathbf{F}}_{\perp} = \frac{1}{\gamma} \mathbf{F}_{\perp}, \quad \bar{\mathbf{F}}_{\parallel} = \mathbf{F}_{\parallel}, \quad (12.68)$$

where $\perp$ means the force is perpendicular to the direction of motion, and $\parallel$ means it is parallel to the direction of motion.

12.2 Relativistic Electrodynamics

If we have a wire carrying a current, with charges moving to the right and left at a speed v , so we have a continuous line charge λ , then we have the current $I = 2\lambda v$. A distnace s away from the wire we have a moving point charge q with velocity u . This means that in the stationary frame ($\mathcal{S}$) there is not electrical force on q . In the moving frame ($\bar{\mathcal{S}}$), where q is stationary, we have a force on q because of the electric field from the wire.

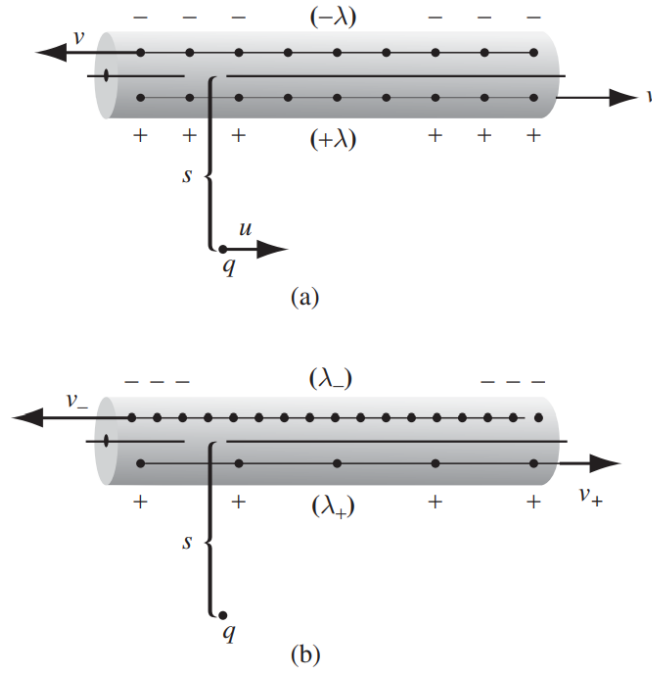


Fig. 12.34

Here the velocities of the charges in the wire are:

$$v_{\pm} = \frac{v \mp u}{1 \mp vu/c^2}, \quad (12.79)$$

but because the velocity of the charges moving in opposite directions are different, the wire is carrying a net negative charge.

$$\lambda_{\pm} = \pm(\gamma_{\pm})\lambda_0, \quad (12.80)$$

where

$$\gamma_{\pm} = \frac{1}{\sqrt{1 - \frac{v_{\pm}^2}{c^2}}}. \quad (12.81)$$

Which means that the net line charge is:

$$\lambda_{tot} = \lambda_+ + \lambda_- = \frac{-2\lambda uv}{c^2 \sqrt{1 - u^2/c^2}}. \quad (12.85)$$

This net line charge means that there is an electric force on the charge in $\bar{\mathcal{S}}$:

$$\bar{F} = qE = q \frac{\lambda_{tot}}{2\pi\epsilon_0 s} = \frac{-q\lambda uv}{\pi\epsilon_0 c^2 s \sqrt{1 - u^2/c^2}}. \quad (12.86)$$

But if there is a force on q in $\bar{\mathcal{S}}$, there must be a force on the wire in $\mathcal{S}$, which we can get by using equation (12.68):

$$F = \frac{1}{\gamma} \bar{F} = \frac{q\lambda uv}{\pi\epsilon_0 c^2 s}, \quad (12.87)$$

this force is a magnetic force.

When a charged object moves, the effect on the electric field comes from length contraction, since the charge density changes. This changes the electric field, only perpendicular to the direction of motion.

$$\sigma = \gamma_0 \sigma_0, \quad (12.92)$$

$$\mathbf{E}_{\perp} = \gamma_0 \mathbf{E}_0, \quad (12.93)$$

$$\mathbf{E}_{\parallel} = \mathbf{E}_0. \quad (12.94)$$

Because of time dialation the magnetic field changes since current is charge per unit time. This means we can write a general transformation of the electric and magnetic fields:

$$\left. \begin{aligned} \bar{E}_x &= E_x, & \bar{E}_y &= \gamma(E_y - vB_z), & \bar{E}_z &= \gamma(E_z + vB_y), \\ \bar{B}_x &= B_x, & \bar{B}_y &= \gamma(B_y + \frac{v}{c^2}E_z), & \bar{B}_z &= \gamma(B_z - \frac{v}{c^2}E_y), \end{aligned} \right\} \quad (12.111)$$

or

$$\left. \begin{aligned} \bar{\mathbf{E}}_{\parallel} &= \mathbf{E}_{\parallel}, & \bar{\mathbf{E}}_{\perp} &= \gamma(\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B}_{\perp}), \\ \bar{\mathbf{B}}_{\parallel} &= \mathbf{B}_{\parallel}, & \bar{\mathbf{B}}_{\perp} &= \gamma(\mathbf{B}_{\perp} - \frac{\mathbf{v}}{c^2} \times \mathbf{E}_{\perp}). \end{aligned} \right\} \quad (12.112)$$

From this we see two simple relations if $\mathbf{B} = 0$ or $\mathbf{E} = 0$ in $\mathcal{S}$, then we have:

$$\bar{\mathbf{B}} = -\frac{1}{c^2}(\mathbf{v} \times \mathbf{E}), \quad (12.113)$$

$$\bar{\mathbf{E}} = \mathbf{v} \times \mathbf{B}. \quad (12.114)$$

Field Tensor

The transformations for the fields do not follow the transformations for the spartial parts. We can therefore not use a normal 4-vector transformation matrix, but instead they transform with a antisymmetric second-rank tensor. A second-rank tensor means that it transform with two factors:

$$\bar{t}^{\mu\nu} = \Lambda_{\lambda}^{\mu} \Lambda_{\sigma}^{\nu} t^{\lambda\sigma}, \quad (12.118)$$

and antisymmetric means that $t^{\mu\nu} = -t^{\nu\mu}$:

$$t^{\mu\nu} = \begin{bmatrix} 0 & t^{01} & t^{02} & t^{03} \\ -t^{01} & 0 & t^{12} & t^{13} \\ -t^{02} & -t^{12} & 0 & t^{23} \\ -t^{03} & -t^{13} & -t^{23} & 0 \end{bmatrix}. \quad (12.120)$$

So using (12.118) we get:

$$\bar{t}^{01} = \Lambda_0^0 \Lambda_0^1 t^{00} + \Lambda_0^0 \Lambda_1^1 t^{01} + \Lambda_1^0 \Lambda_0^1 t^{10} + \Lambda_1^0 \Lambda_1^1 t^{11},$$

which can be done for all parts of the matrix, giving us the final transformations:

$$\left. \begin{aligned} \bar{t}^{01} &= t^{01}, & \bar{t}^{02} &= \gamma(t^{02} - \beta t^{12}), & \bar{t}^{03} &= \gamma(t^{03} + \beta t^{13}), \\ \bar{t}^{23} &= t^{23}, & \bar{t}^{13} &= \gamma(t^{13} + \beta t^{03}), & \bar{t}^{12} &= \gamma(t^{12} - \beta t^{02}). \end{aligned} \right\} \quad (12.121)$$

With these transformation rules we get the field tensor:

$$F^{\mu\nu} = \begin{bmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & B_z & B_y \\ -E_y/c & -B_z & 0 & B_x \\ -E_z/c & B_y & -B_x & 0 \end{bmatrix}, \quad (12.122)$$

or you can write it as the dual tensor which gives the same transformations:

$$G^{\mu\nu} = \begin{bmatrix} 0 & B_x & B_y & B_z \\ -B_x & 0 & -E_z/c & E_y/c \\ -B_y & E_z/c & 0 & -E_x/c \\ -B_z & -E_y/c & E_x/c & 0 \end{bmatrix}. \quad (12.123)$$

With these tensors we can rewrite everything we have learnt so far.

$$\rho = \gamma \rho_0, \quad \mathbf{J} = \mathbf{v} \gamma \rho_0, \quad (12.124)$$

we make charge and current density a 4-vector:

$$J^{\mu} = \rho_0 \eta^{\mu} = (cp, J_x, J_y, J_z), \quad (12.127)$$

which is divergenless:

$$\frac{\partial J^{\mu}}{\partial x^{\mu}} = 0. \quad (12.129)$$

Maxwell's equations can be written as:

$$\frac{\partial F^{\mu\nu}}{\partial x^{\nu}} = \mu_0 J^{\mu}, \quad \frac{\partial G^{\mu\nu}}{\partial x^{\nu}} = 0. \quad (12.130)$$

with summation over ν . The Minkowski force on a charge q is:

$$K^\mu = q\eta_\nu F^{\mu\nu}. \quad (12.131)$$

We can also express the potentials in 4-vector notation.

$$A^\mu = (V/c, A_x, A_y, A_z), \quad (12.135)$$

$$F^{\mu\nu} = \frac{\partial A^\nu}{\partial x_\mu} - \frac{\partial A^\mu}{\partial x_\nu}. \quad (12.136)$$

Remember that the subscript means that $x_0 = -ct$. Gauge transformations also work:

$$A^{\mu'} = A^\mu + \frac{\partial \lambda}{\partial x^\mu}, \quad (12.138)$$

we can therefore use the Lorentz gauge and get the ultimate electrodynamic equation:

$$\square^2 A^\mu = -\mu_0 J^\mu, \quad (12.140)$$

where

$$\square^2 = \frac{\partial}{\partial x_\nu} \frac{\partial}{\partial x^\nu} = \nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}. \quad (12.141)$$

13 Appendix

Tables

Table 4.2. Dielectric constants (unless otherwise specified, values given are for 1 atm, 20°C). Data from *Handbook of Chemistry and Physics*, 91st ed., CRC Press, Boca Raton (2010).

Material	Dielectric constant	Material	Dielectric constant
Vacuum	1	Benzene	2.28
Helium	1.000065	Diamond	5.7–5.9
Neon	1.00013	Sodium chloride	5.9
Hydrogen (H ₂)	1.000254	Silicon	11.7
Argon	1.000517	Methanol	33.0
Air (dry)	1.000536	Water	80.1
Nitrogen (N ₂)	1.000548	Ice (−30°C)	104
Water vapor (100°C)	1.00589	KTaNbO ₃ (0°C)	34,000

Table 6.1. Magnetic susceptibilities (unless otherwise specified, values are for 1 atm, 20°C). Data from *Handbook of Chemistry and Physics*, 91st ed., CRC Press, Boca Raton (2010) and other references.

Material	Susceptibility	Material	Susceptibility
<i>Diamagnetic:</i>		<i>Paramagnetic:</i>	
Bismuth	-1.7×10^{-4}	Oxygen (O ₂)	1.7×10^{-6}
Gold	-3.4×10^{-5}	Sodium	8.5×10^{-6}
Silver	-2.4×10^{-5}	Aluminum	2.2×10^{-5}
Copper	-9.7×10^{-6}	Tungsten	7.0×10^{-5}
Water	-9.0×10^{-6}	Platinum	2.7×10^{-4}
Carbon dioxide	-1.1×10^{-8}	Oxygen (liquid) (−200°C)	3.9×10^{-3}
Hydrogen (H ₂)	-2.1×10^{-9}	Gadolinium	4.8×10^{-1}

Table 7.1. Resistivities, in ohm meters (all values are for 1 atm, 20°C). Data from *Handbook of Chemistry and Physics*, 91st ed., CRC Press, Boca Raton (2010) and other references.

Material	Resistivity	Material	Resistivity
<i>Conductors:</i>		<i>Semiconductors:</i>	
Silver	1.59×10^{-8}	Sea water	0.2
Copper	1.68×10^{-8}	Germanium	0.46
Gold	2.21×10^{-8}	Diamond	2.7
Aluminum	2.65×10^{-8}	Silicon	2500
Iron	9.61×10^{-8}	<i>Insulators:</i>	
Mercury	9.61×10^{-7}	Water (pure)	8.3×10^3
Nichrome	1.08×10^{-6}	Glass	10^9 – 10^{14}
Manganese	1.44×10^{-6}	Rubber	10^{13} – 10^{15}
Graphite	1.6×10^{-5}	Teflon	10^{22} – 10^{24}

14 To Do

- Something about direction of induced electric field or emf
- section 11.1.4 if covered in lecture 23